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Chu et al.

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(54) **SEMICONDUCTOR DEVICE STRUCTURE
WITH BACKSIDE CONTACT**

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H01L 21/84 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 30/6729** (2025.01); **H10D 30/031**
(2025.01); **H10D 86/00** (2025.01);
(Continued)

(58) **Field of Classification Search**

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H10D 86/01; H10D 30/6735; H10D
30/6737; H10D 30/6743; H10D 30/6757;
H01L 21/76897; H01L 23/5286; H01L
29/1079; H01L 29/0673; H01L 29/0847;
H01L 29/66439; H01L 29/41725; H01L
29/775; H01L 29/41733; H01L 29/42392;
H01L 29/78696; B82Y 10/00
(Continued)

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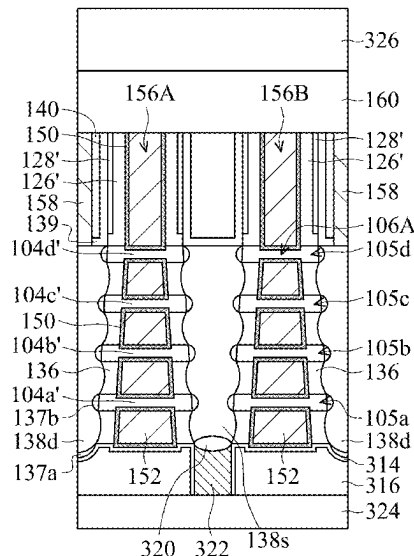
Primary Examiner — Tong-Ho Kim

(74) *Attorney, Agent, or Firm* — Birch, Stewart, Kolasch
& Birch, LLP

(57) **ABSTRACT**

A semiconductor device structure is provided. The semicon-
ductor device structure includes a stack of channel structures
and a first epitaxial structure and a second epitaxial structure
adjacent to opposite sides of the channel structures. The
semiconductor device structure also includes a gate stack
wrapped around the channel structures and a backside
conductive contact connected to the second epitaxial struc-
ture. The second epitaxial structure is between a top of the
backside conductive contact and a top of the gate stack. The
semiconductor device structure further includes an etch stop
layer extending along a sidewall of the backside conductive
contact and a bottom of the gate stack.

20 Claims, 33 Drawing Sheets



(51)	Int. Cl. <i>H01L 27/12</i> (2006.01) <i>H01L 29/423</i> (2006.01) <i>H01L 29/45</i> (2006.01) <i>H01L 29/66</i> (2006.01) <i>H01L 29/786</i> (2006.01) <i>H10D 30/01</i> (2025.01) <i>H10D 30/67</i> (2025.01) <i>H10D 86/00</i> (2025.01) <i>H10D 86/01</i> (2025.01)	(56)	References Cited U.S. PATENT DOCUMENTS 9,412,817 B2 8/2016 Yang et al. 9,412,828 B2 8/2016 Ching et al. 9,472,618 B2 10/2016 Oxland 9,502,265 B1 11/2016 Jiang et al. 9,520,482 B1 12/2016 Chang et al. 9,536,738 B2 1/2017 Huang et al. 9,576,814 B2 2/2017 Wu et al. 9,608,116 B2 3/2017 Ching et al. 12,015,060 B2 * 6/2024 Chu H01L 29/66439 2021/0399109 A1 12/2021 Su et al. 2021/0408246 A1 12/2021 Ganguly et al. 2022/0139911 A1 * 5/2022 Wei H01L 23/5226 257/365 2022/0165856 A1 5/2022 Yu et al. 2022/0199624 A1 6/2022 Huang et al.
(52)	U.S. Cl. CPC <i>H10D 86/01</i> (2025.01); <i>H10D 30/6735</i> (2025.01); <i>H10D 30/6737</i> (2025.01); <i>H10D</i> <i>30/6743</i> (2025.01); <i>H10D 30/6757</i> (2025.01)		
(58)	Field of Classification Search USPC 257/401 See application file for complete search history.		* cited by examiner

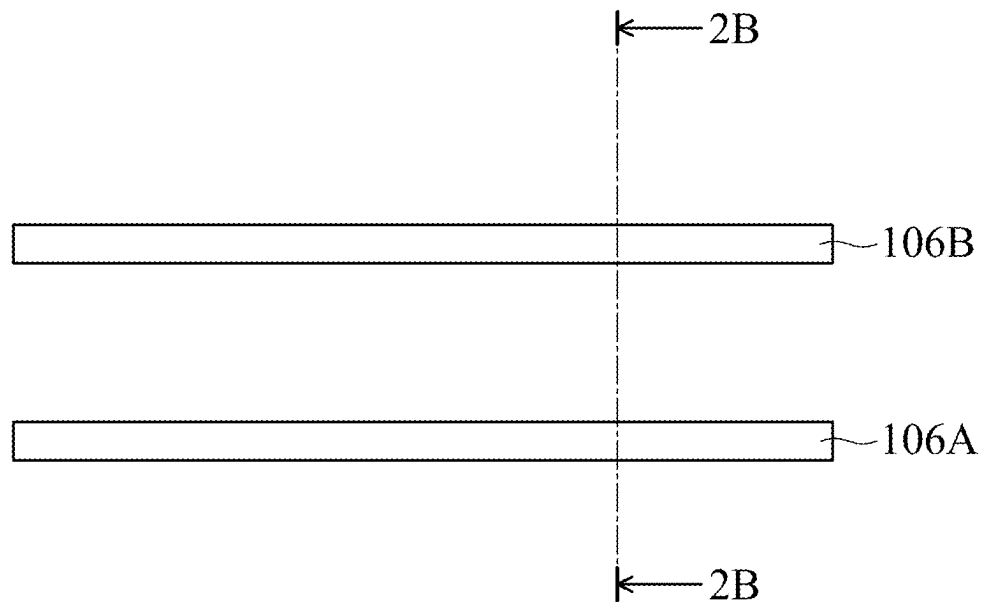


FIG. 1A

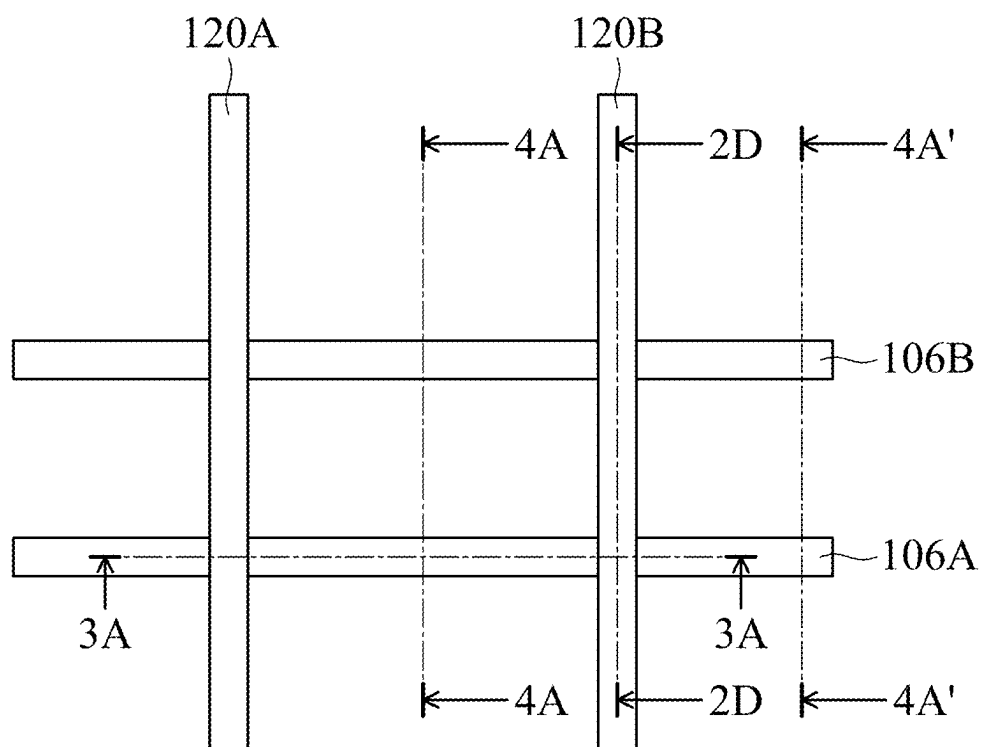


FIG. 1B

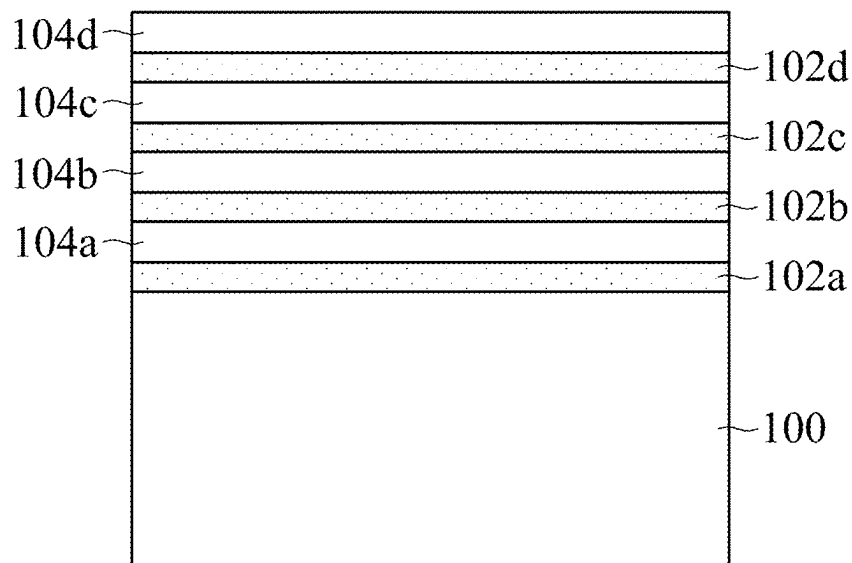


FIG. 2A

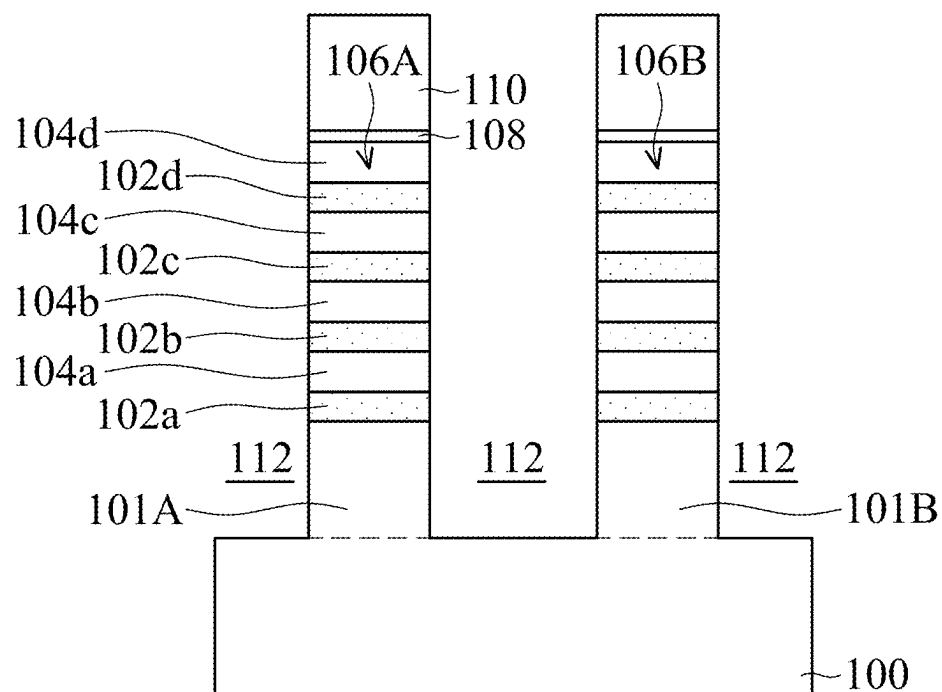


FIG. 2B

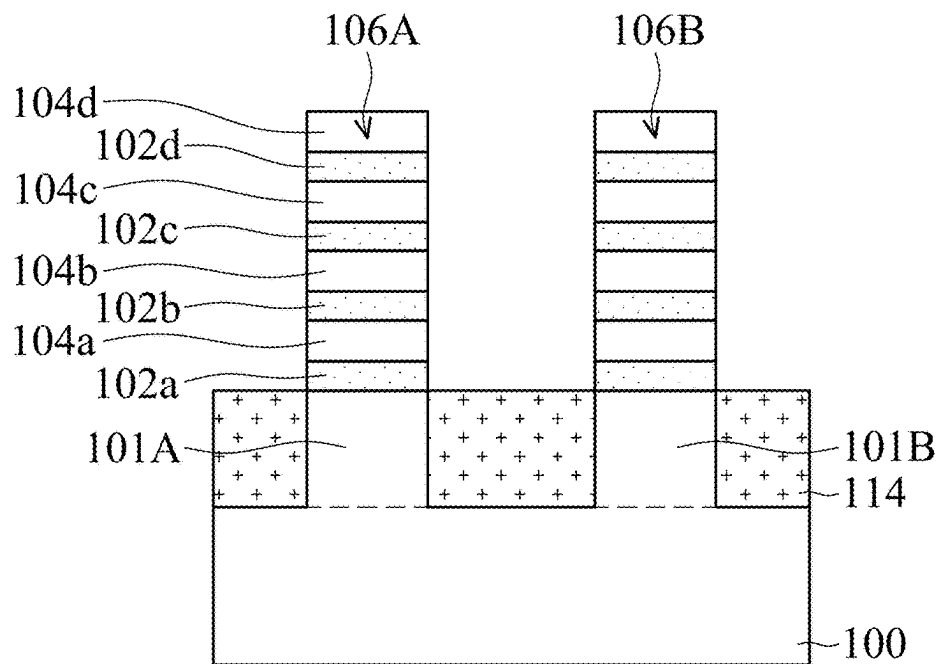


FIG. 2C

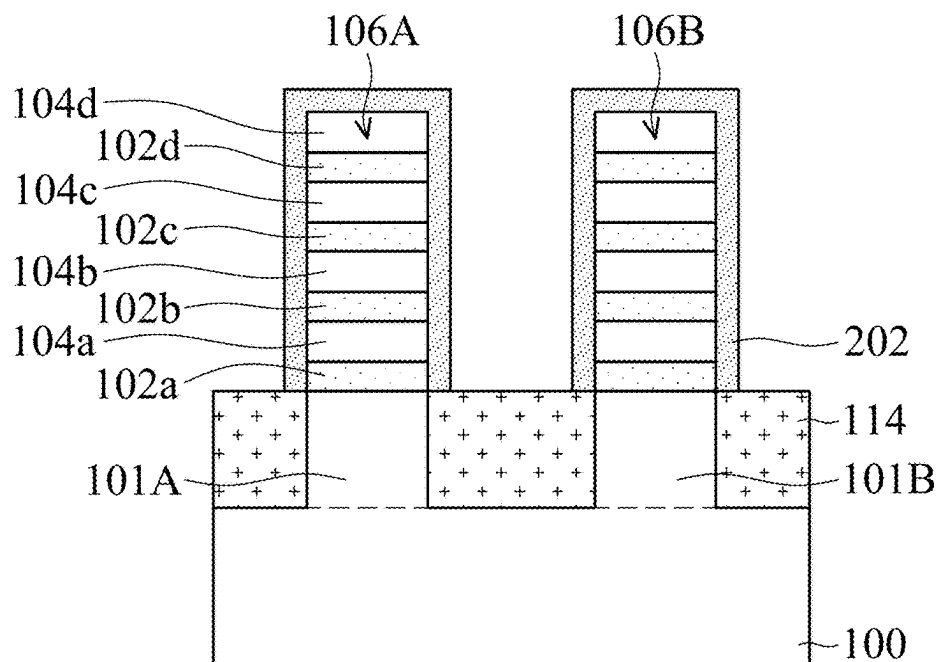


FIG. 2D

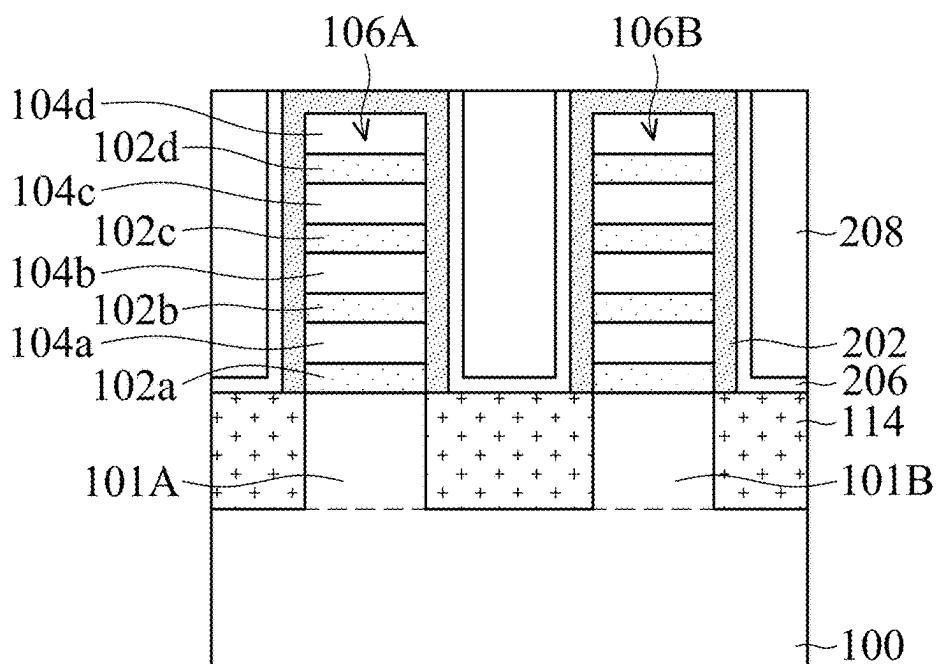


FIG. 2E

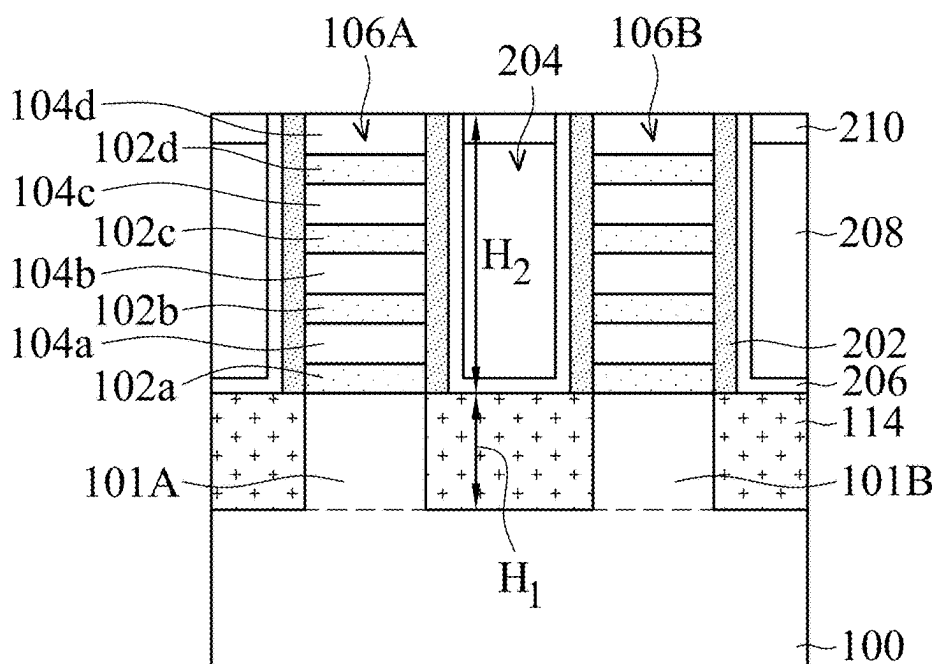


FIG. 2F

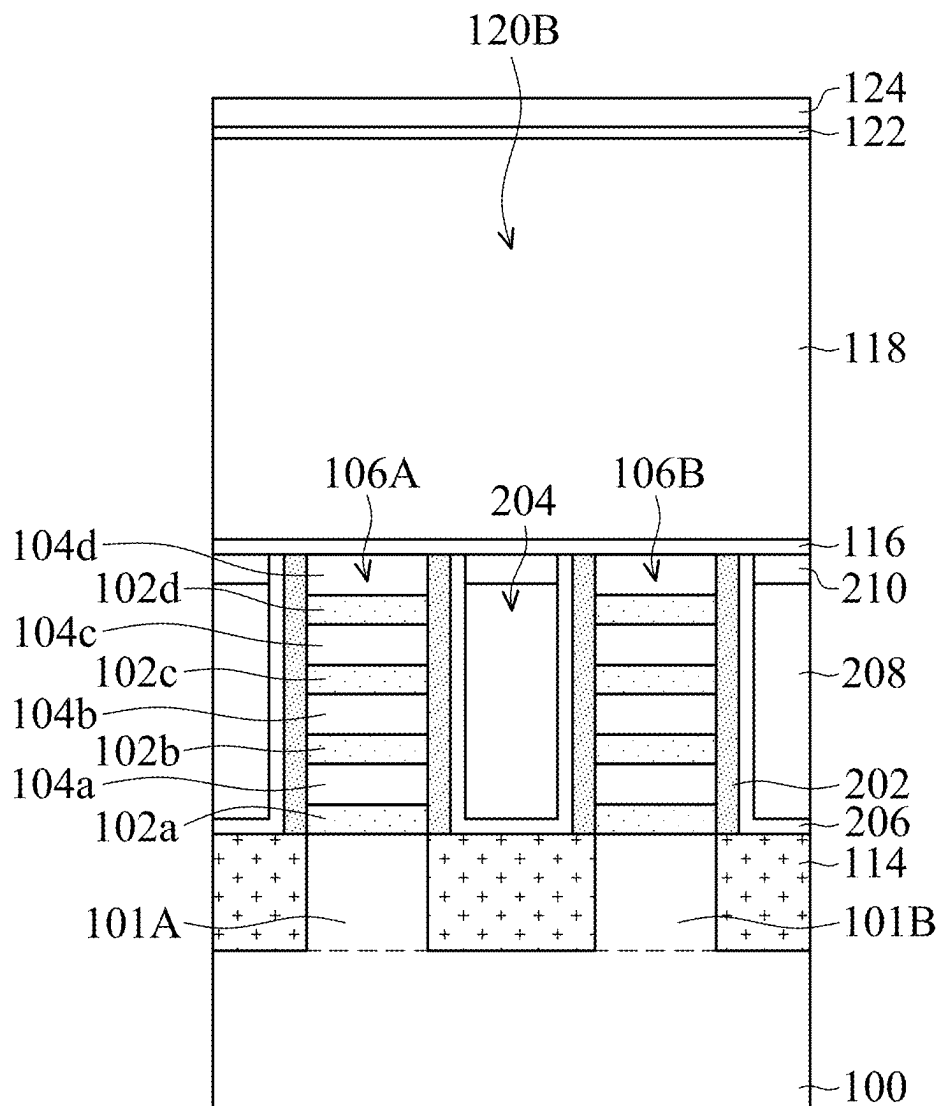


FIG. 2G

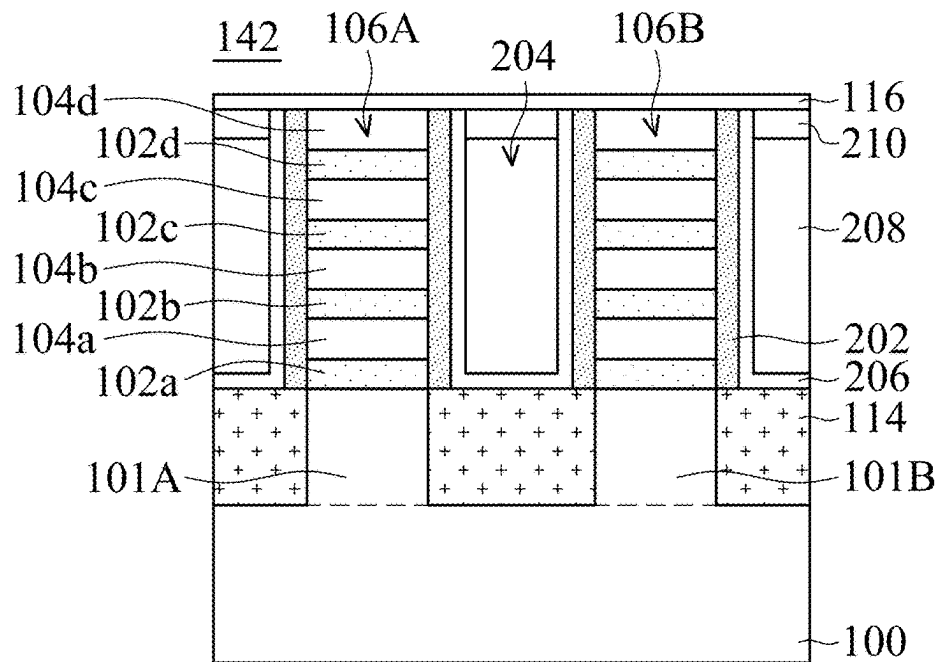


FIG. 2H

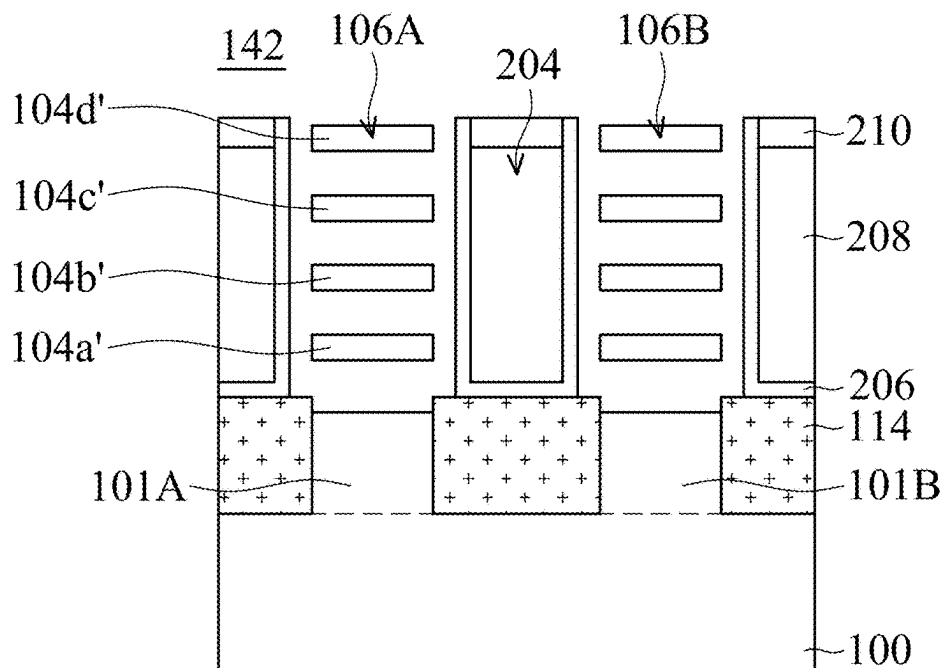


FIG. 2I

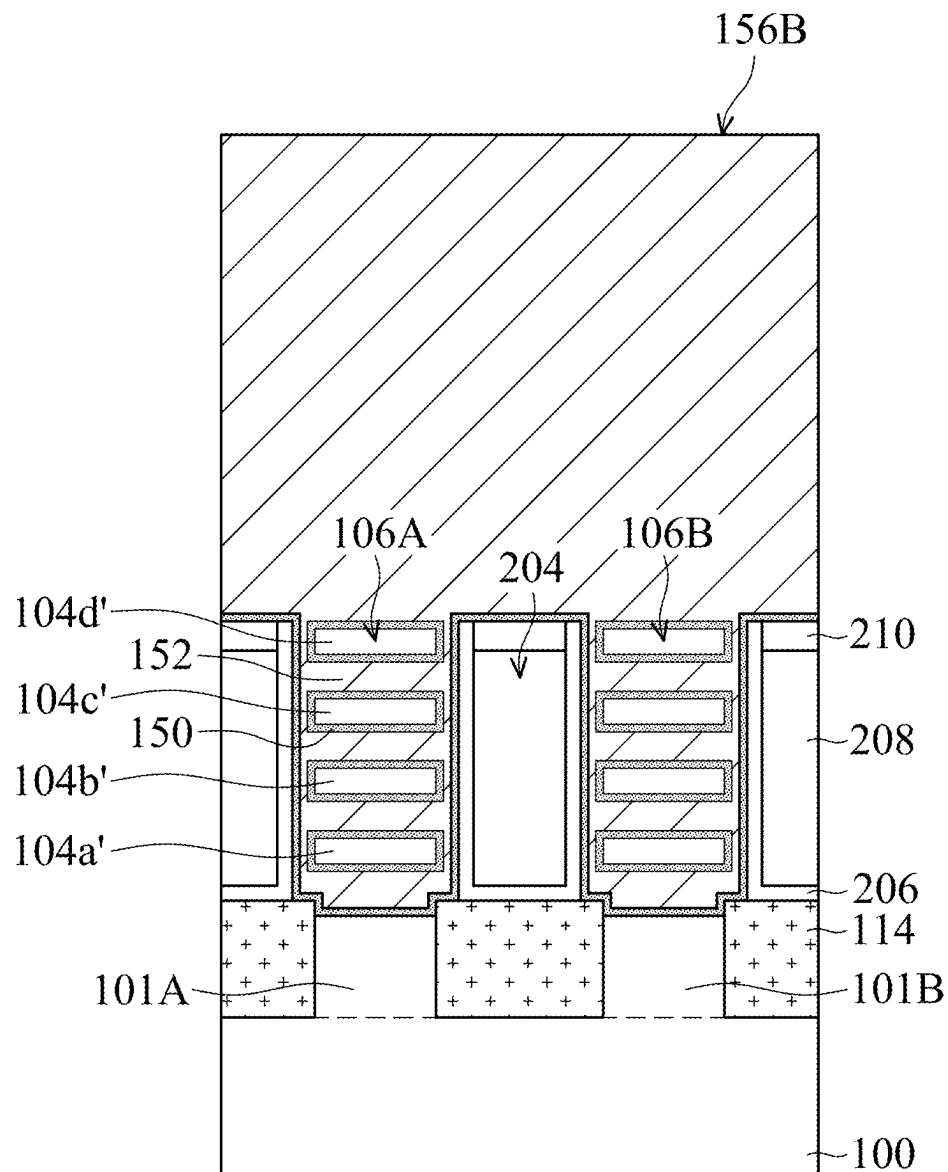


FIG. 2J

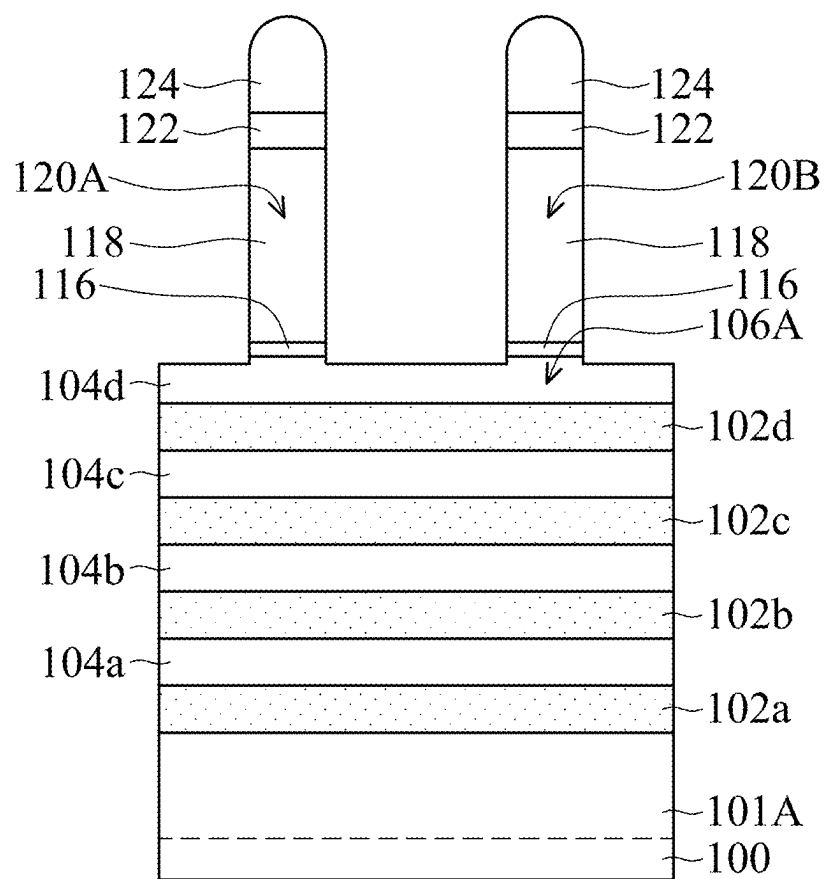


FIG. 3A

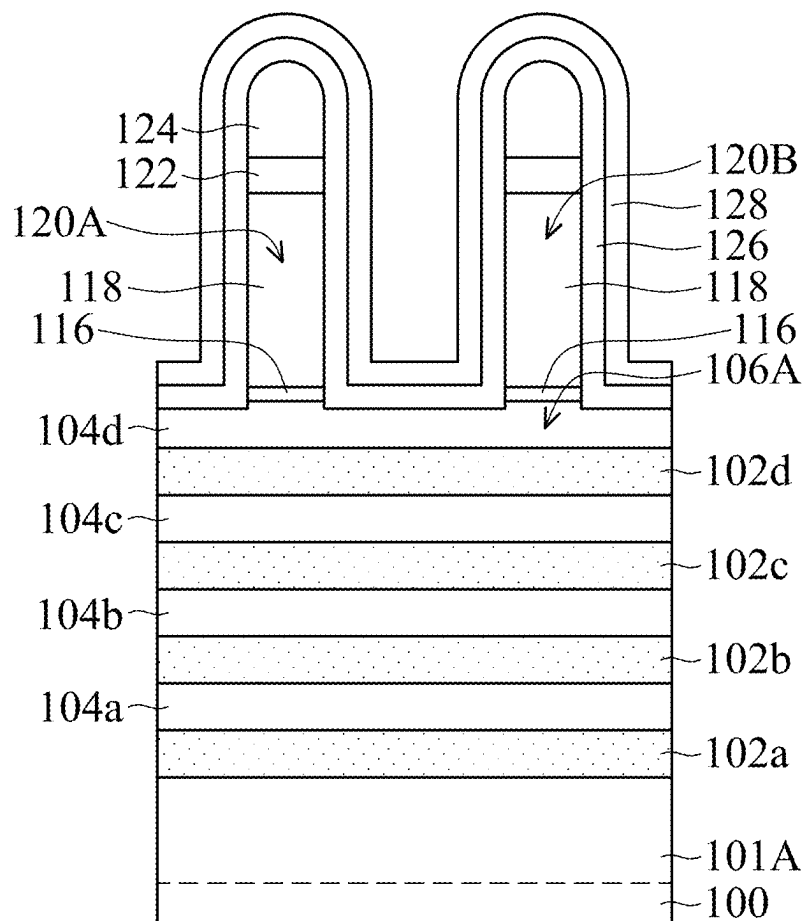


FIG. 3B

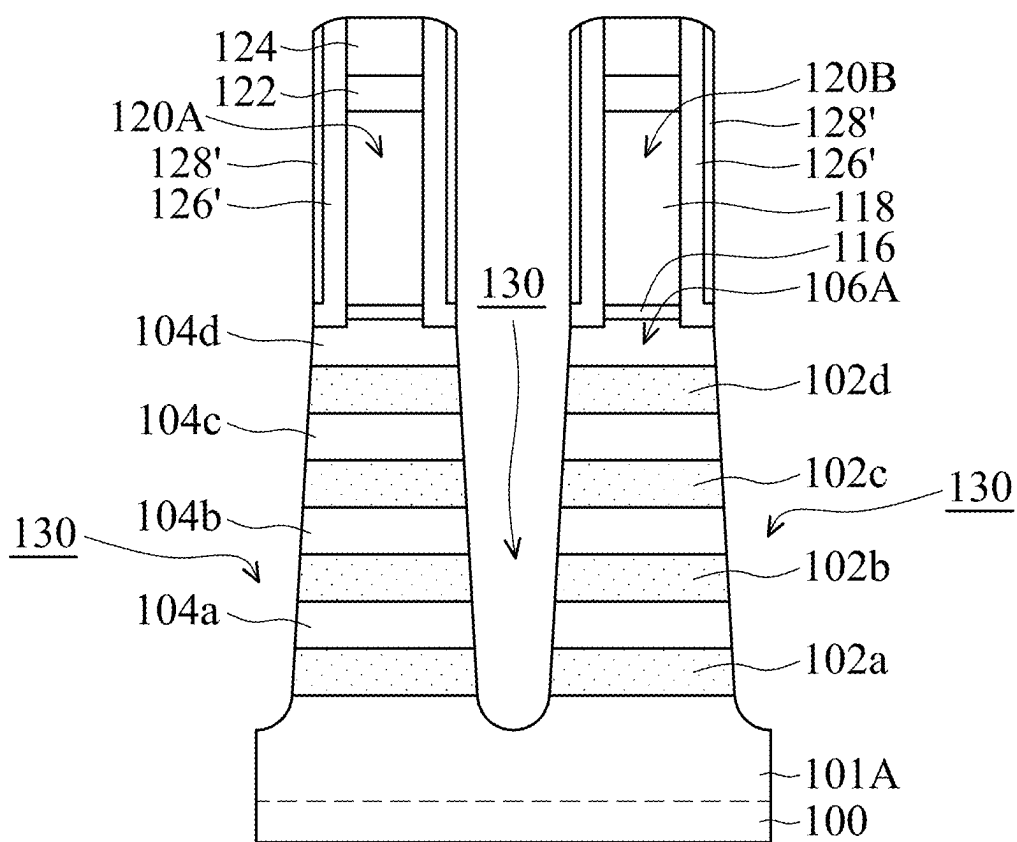


FIG. 3C

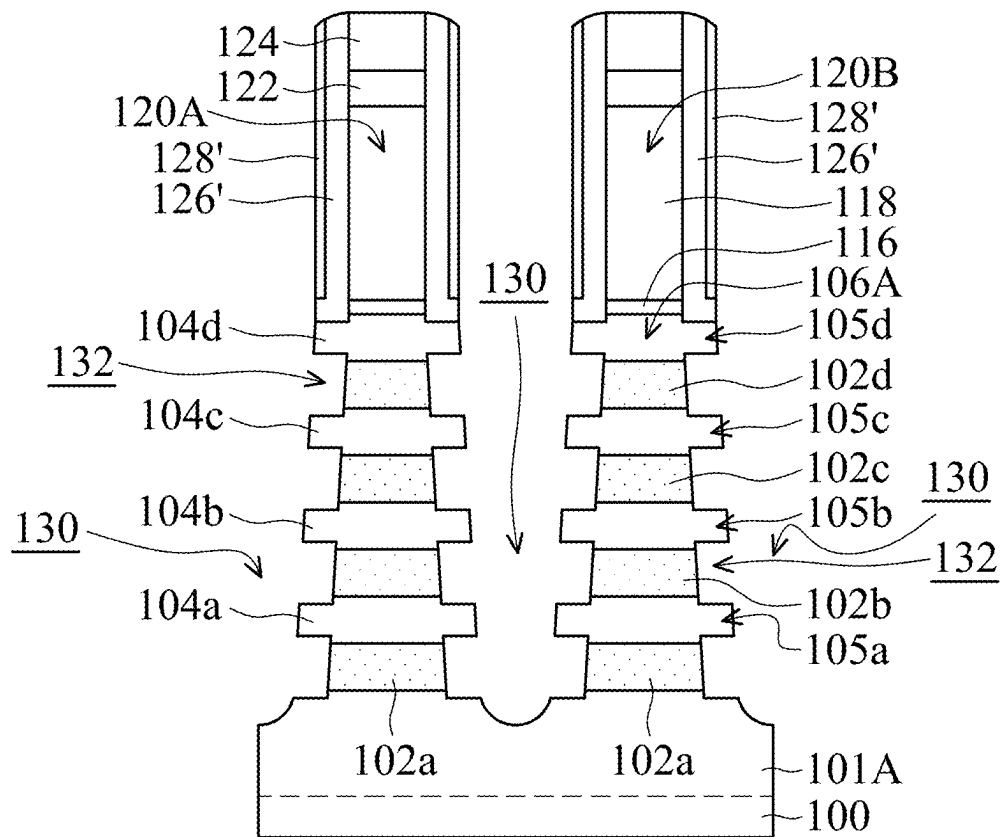


FIG. 3D

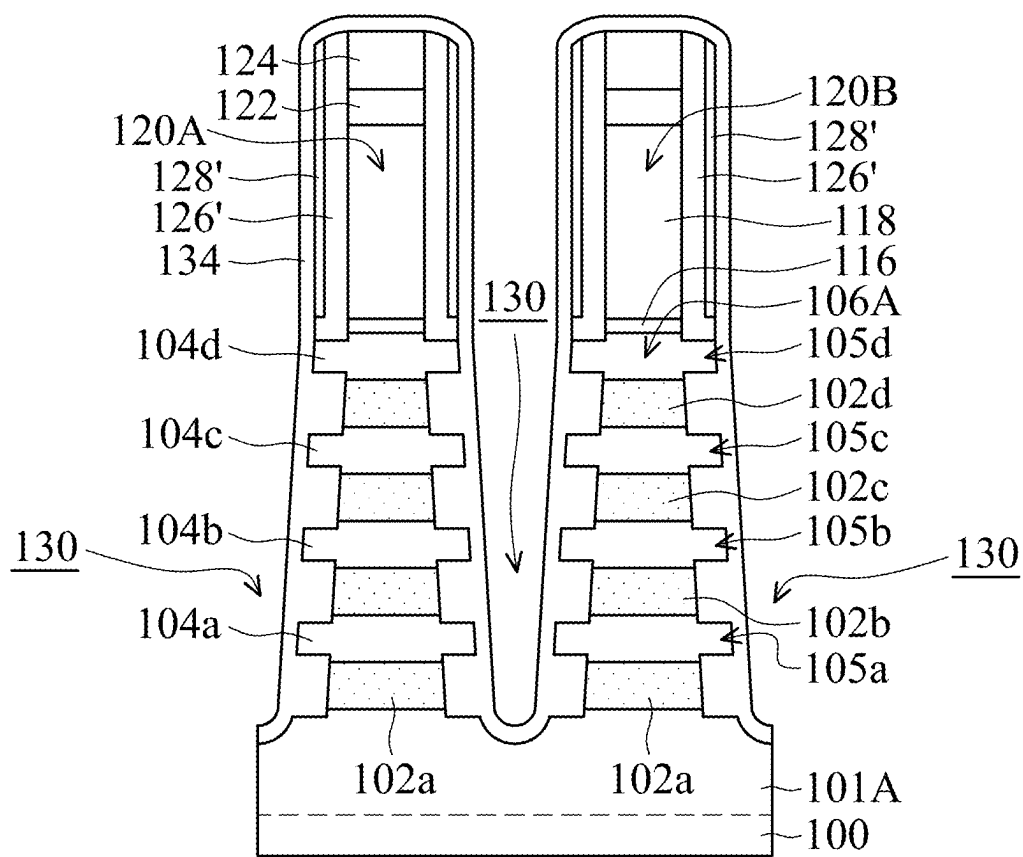


FIG. 3E

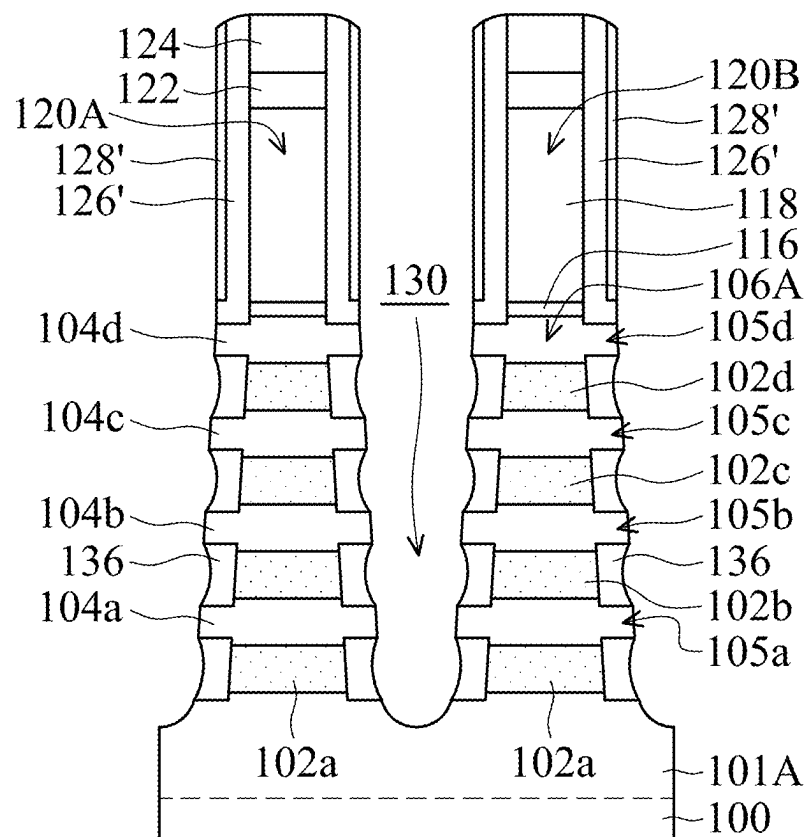


FIG. 3F

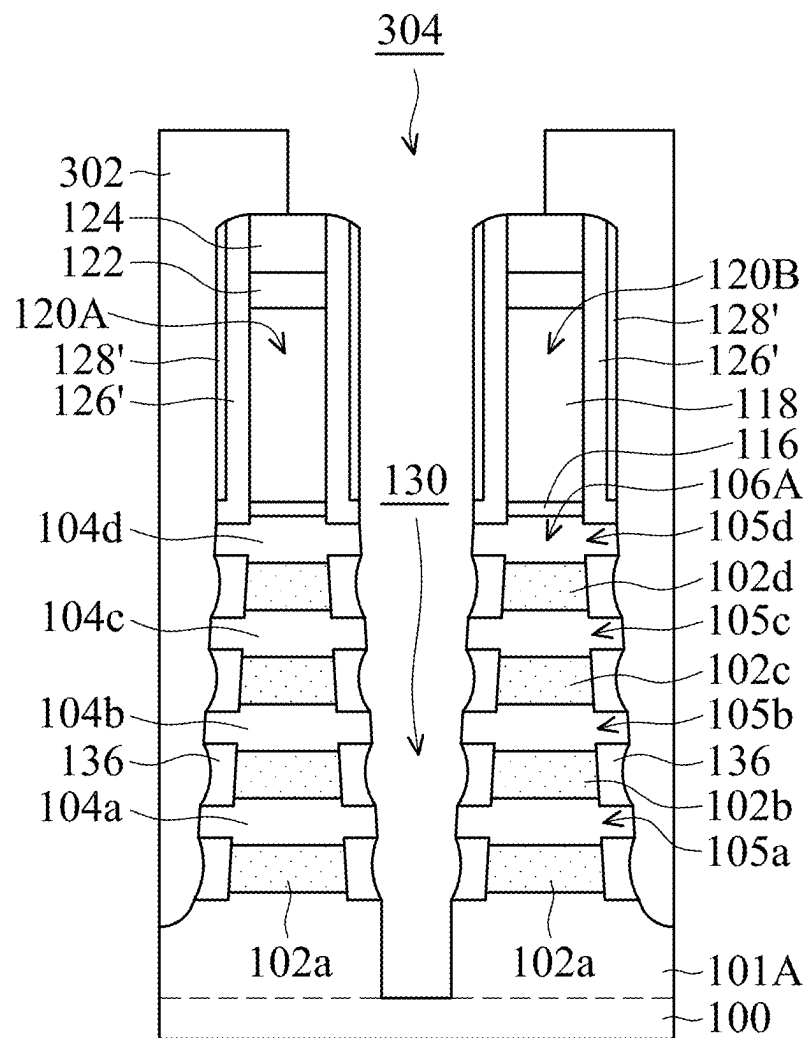


FIG. 3G

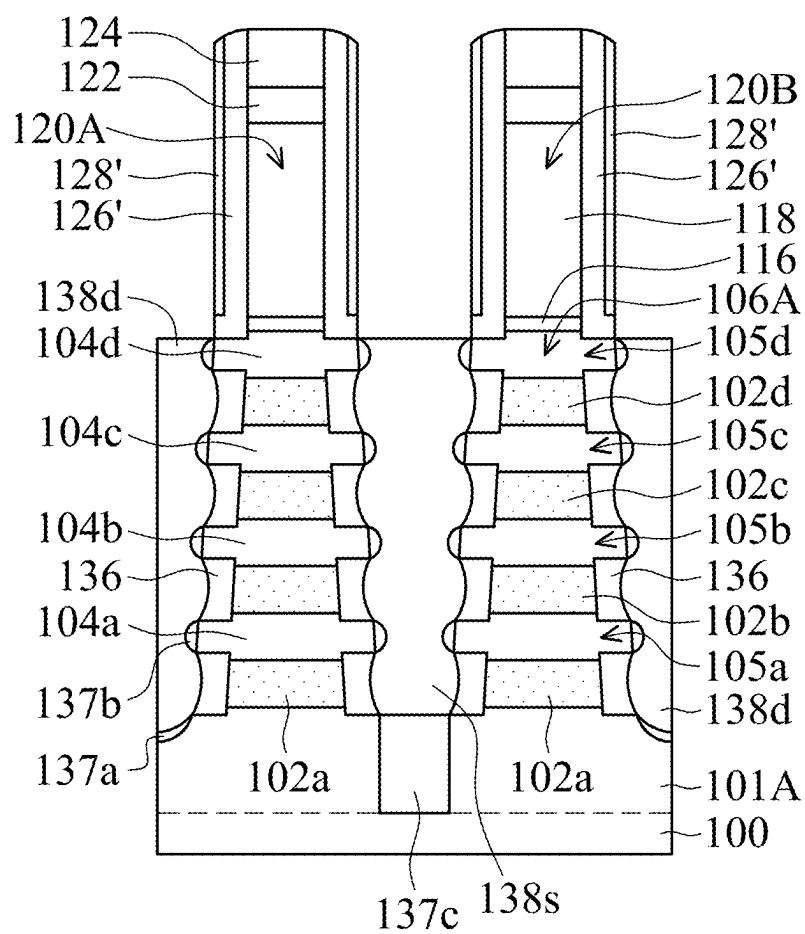


FIG. 3H

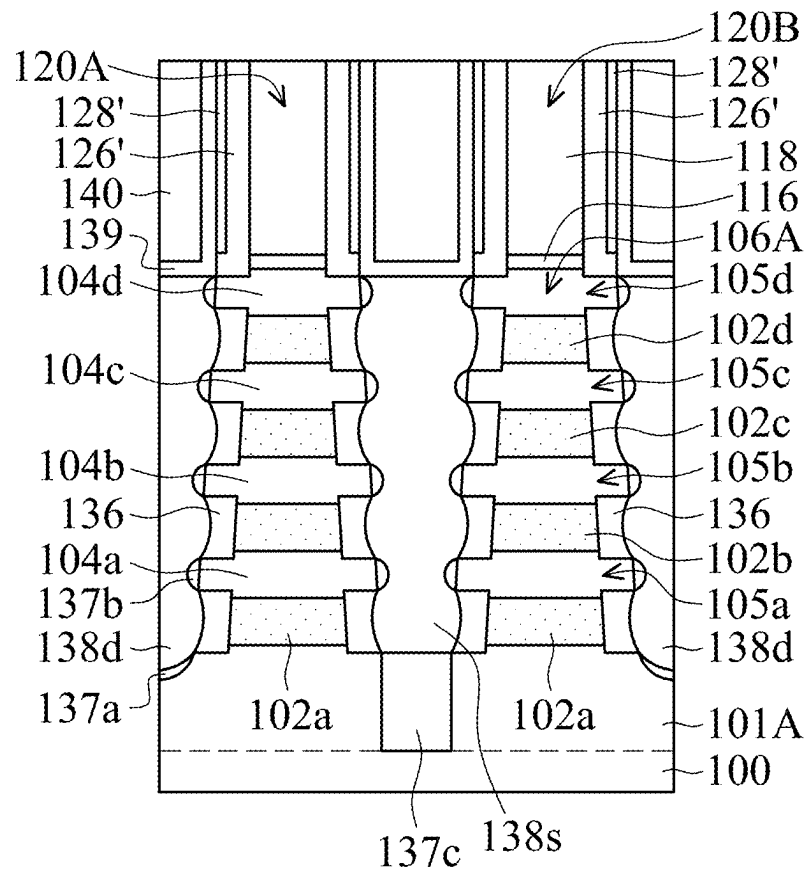


FIG. 3I

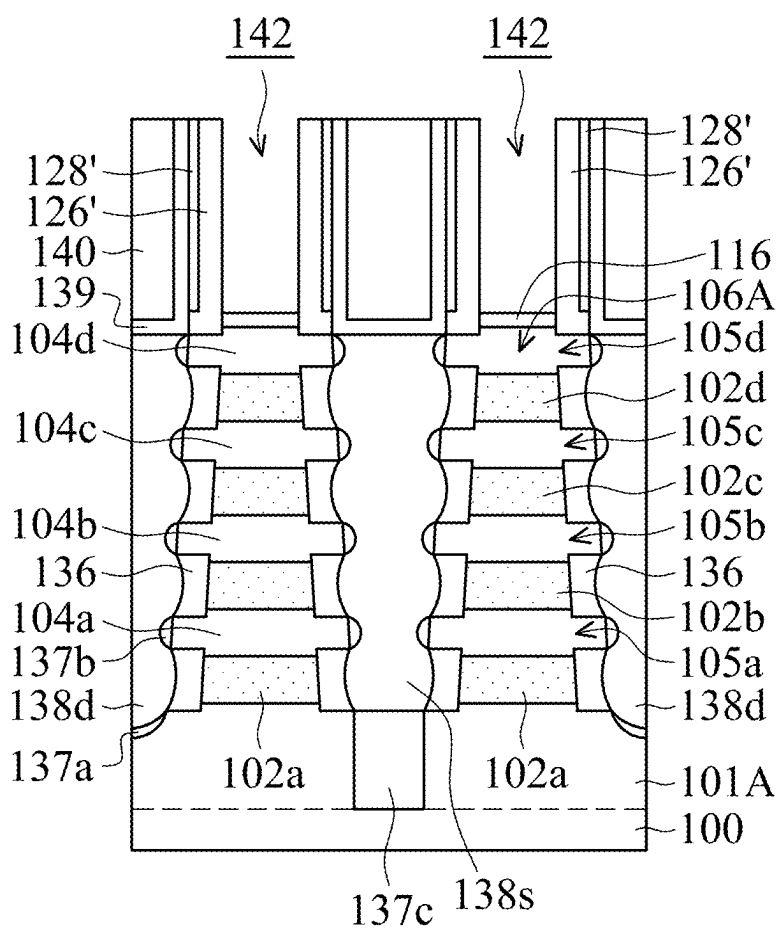


FIG. 3J

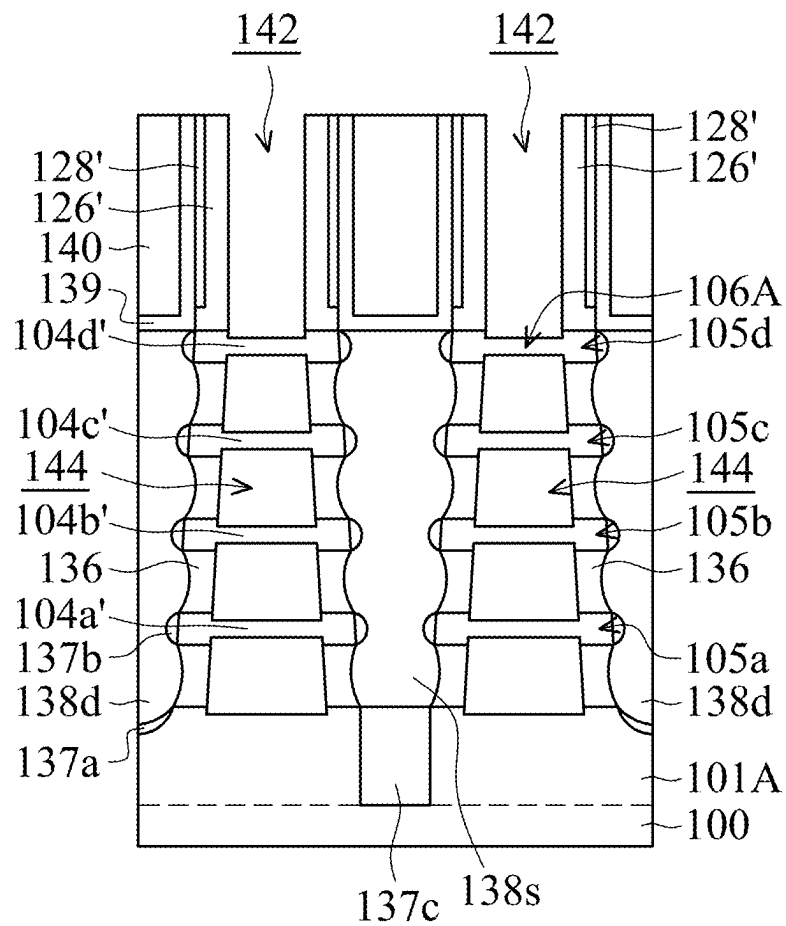


FIG. 3K

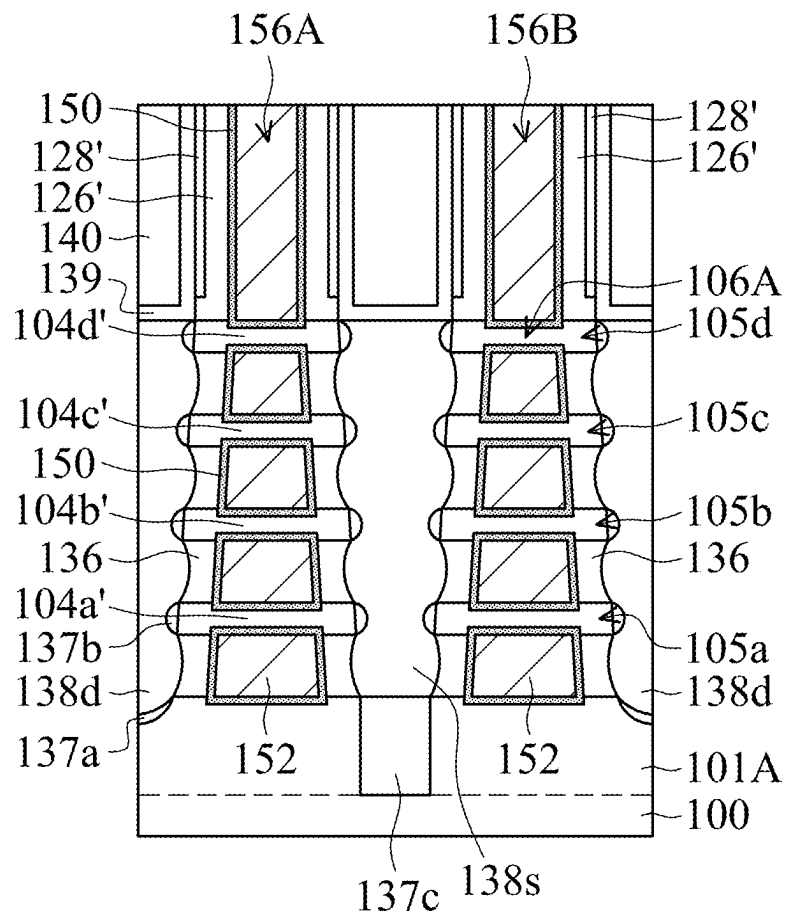


FIG. 3L

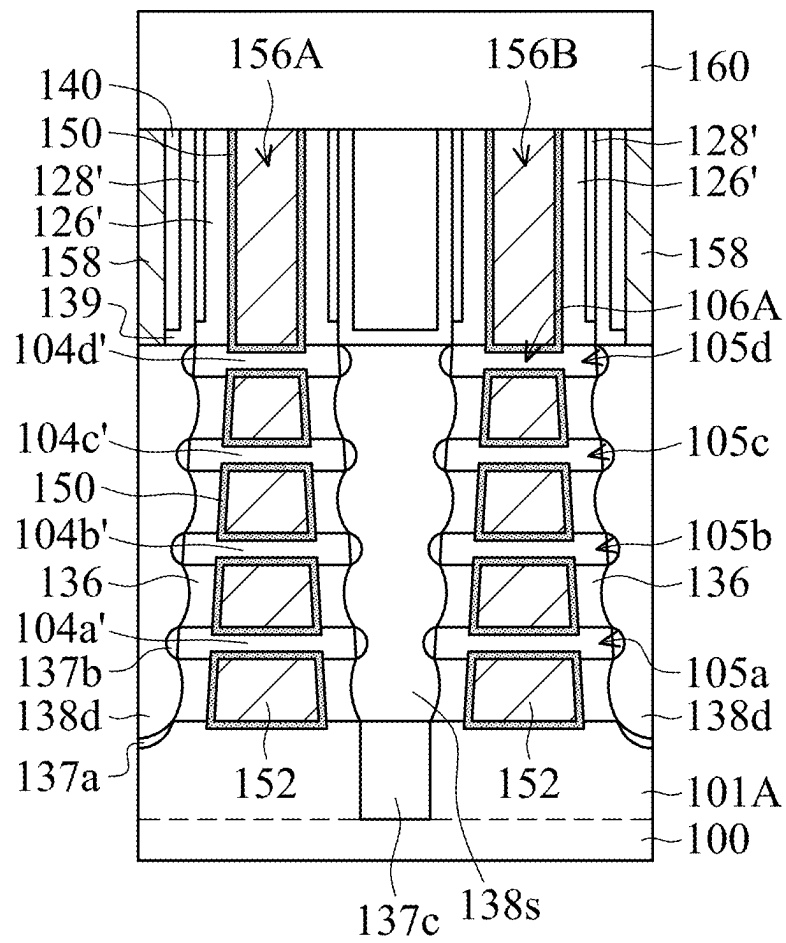


FIG. 3M

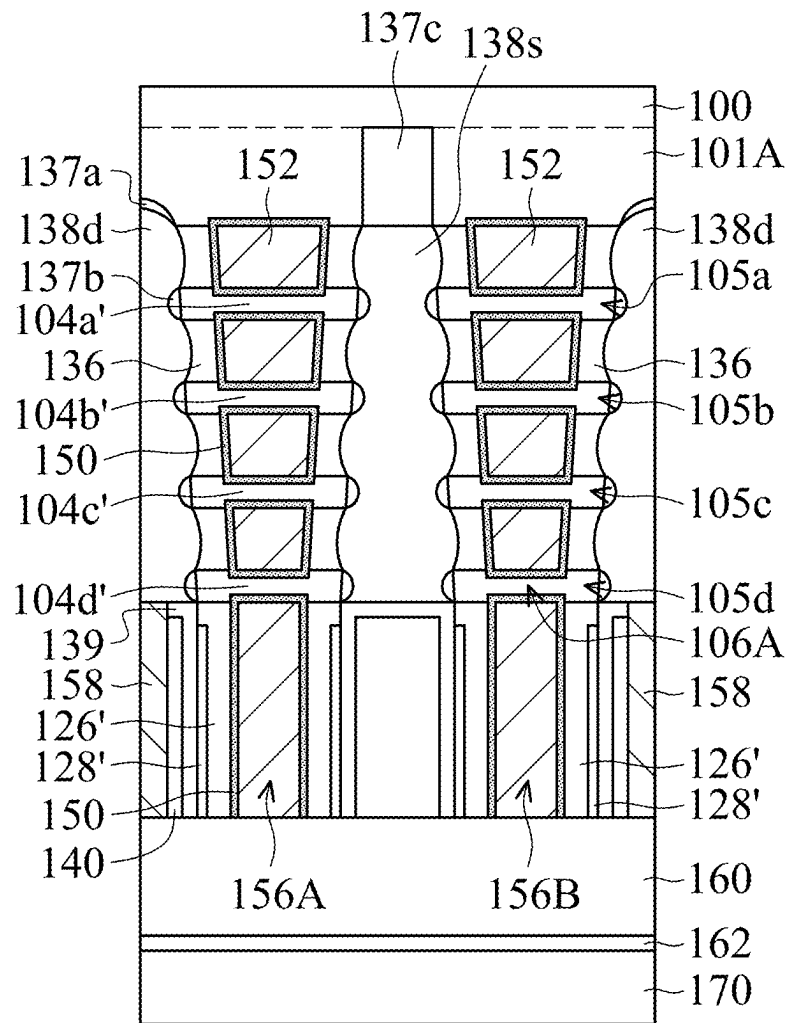


FIG. 3N

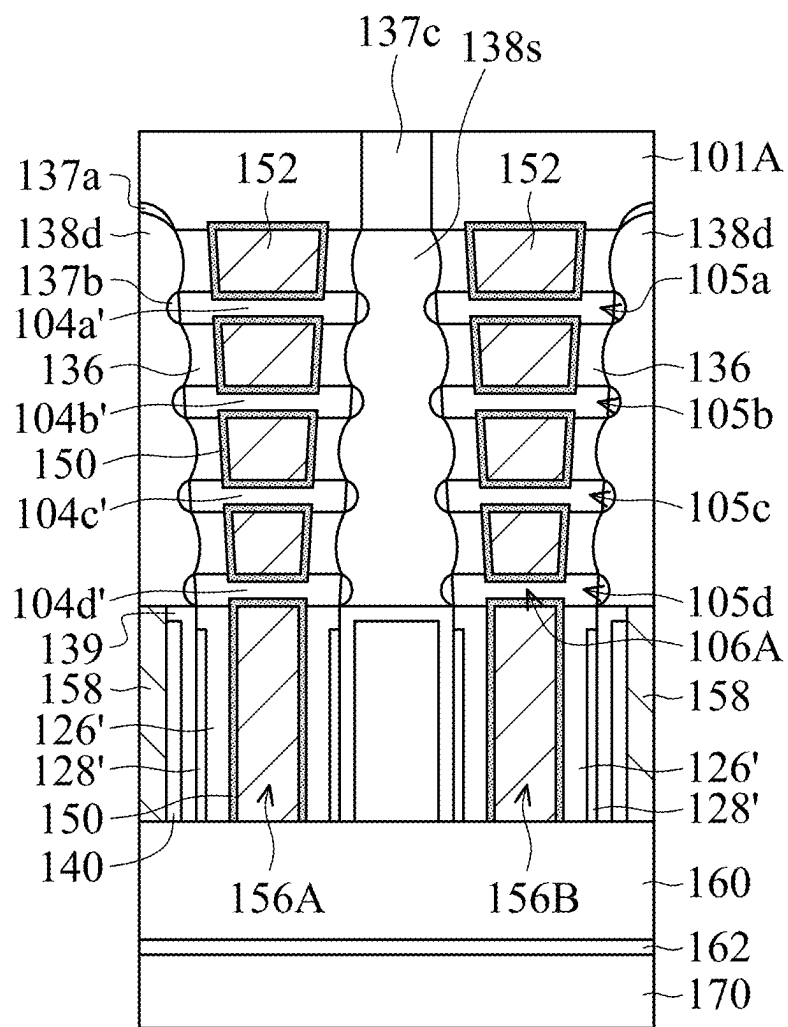


FIG. 30

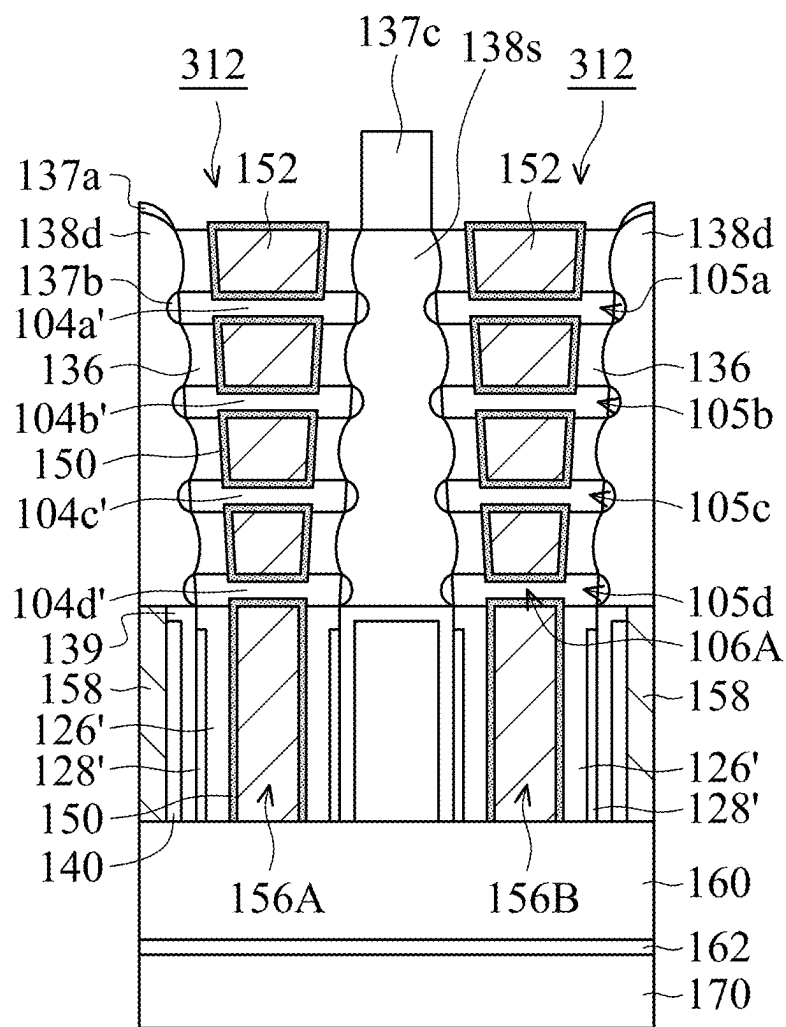


FIG. 3P

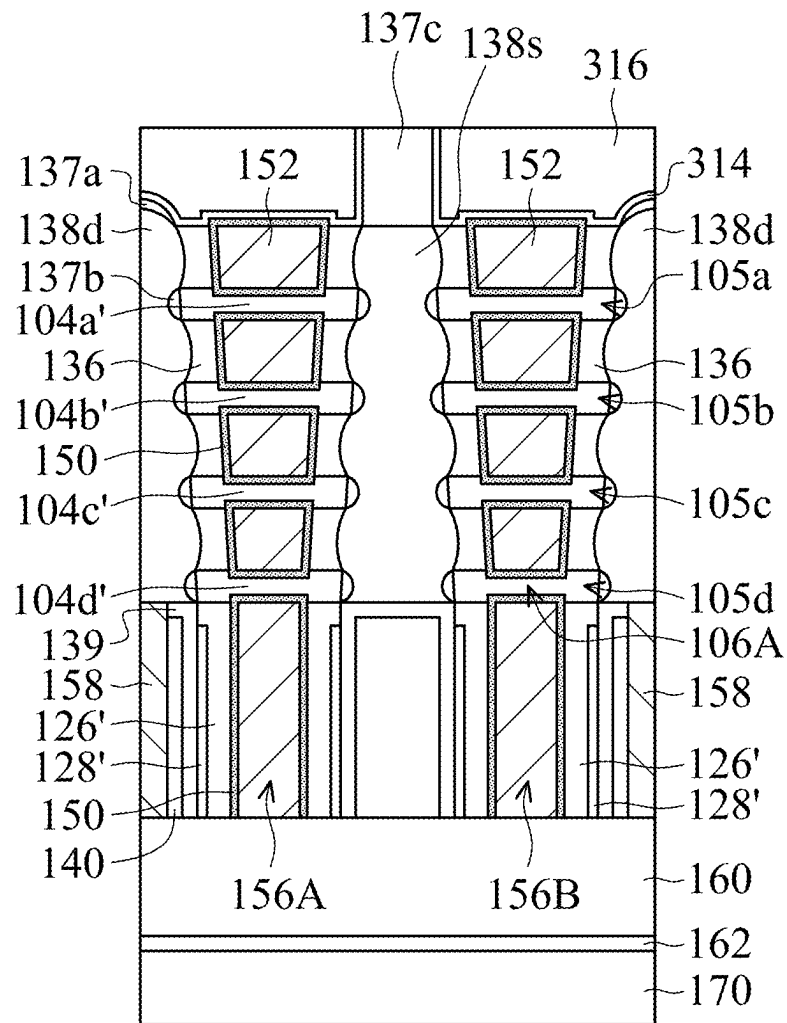


FIG. 3Q

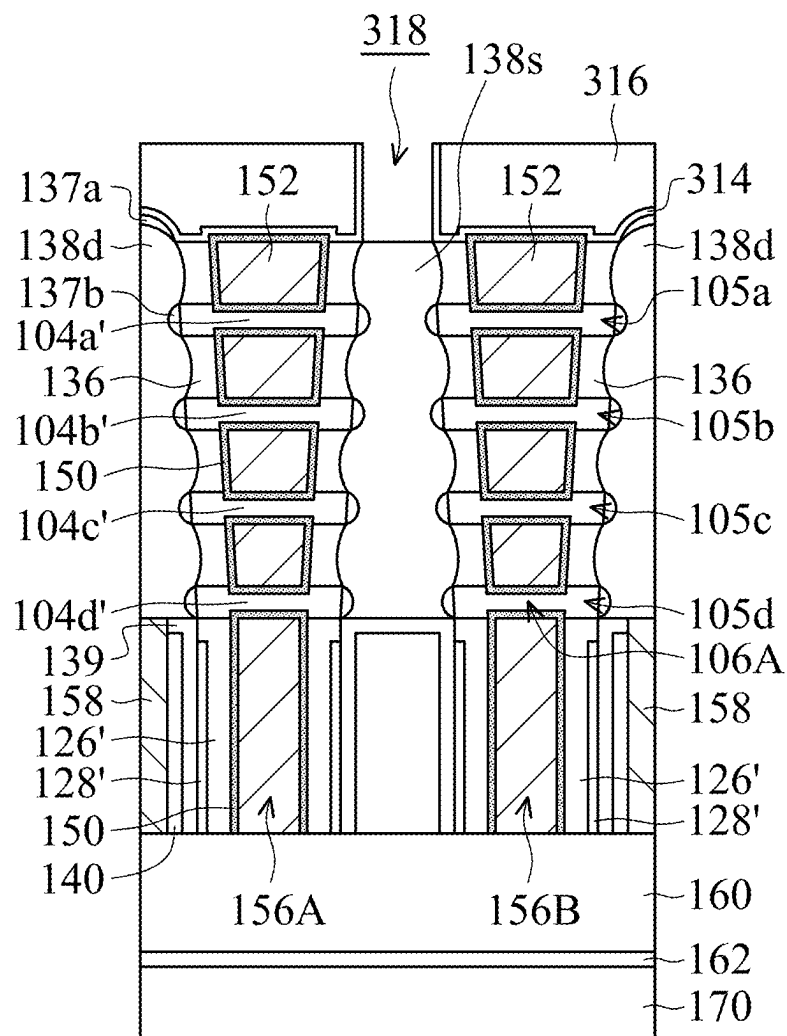


FIG. 3R

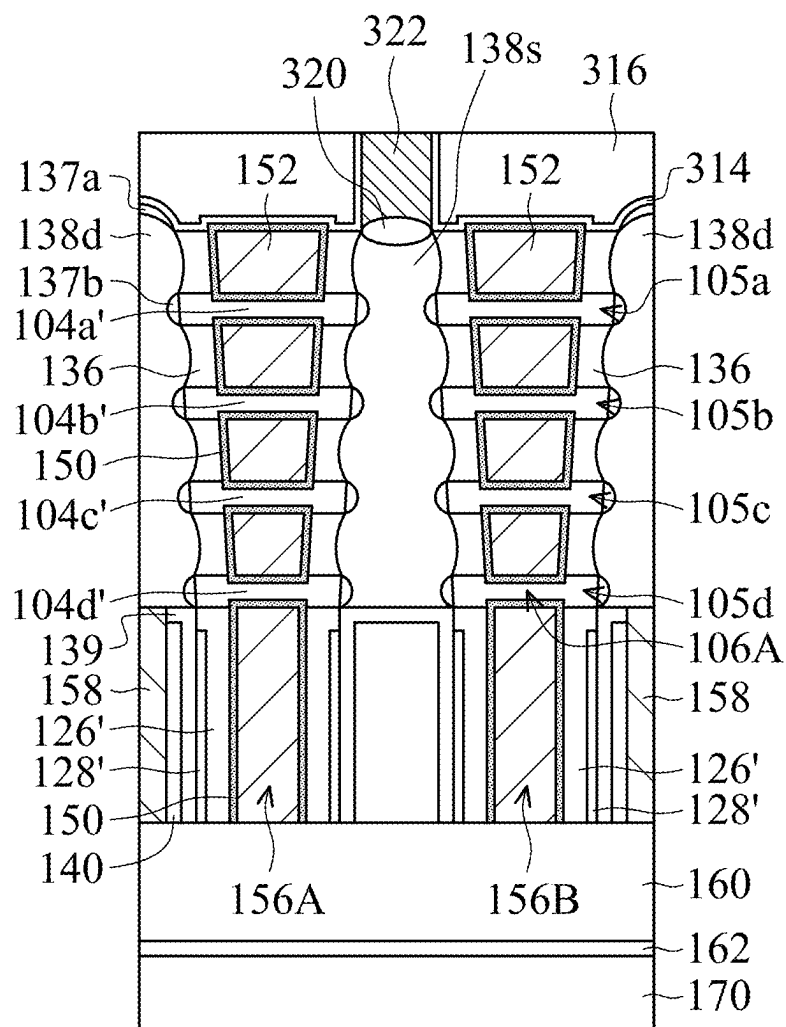


FIG. 3S

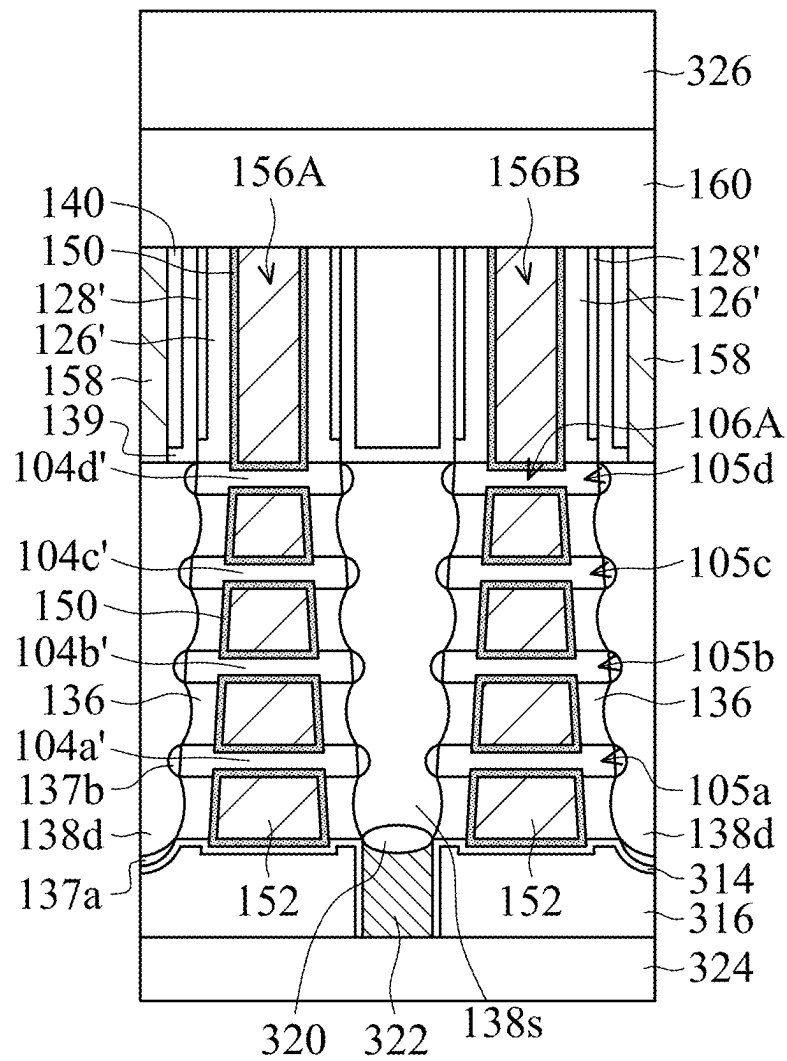


FIG. 3T

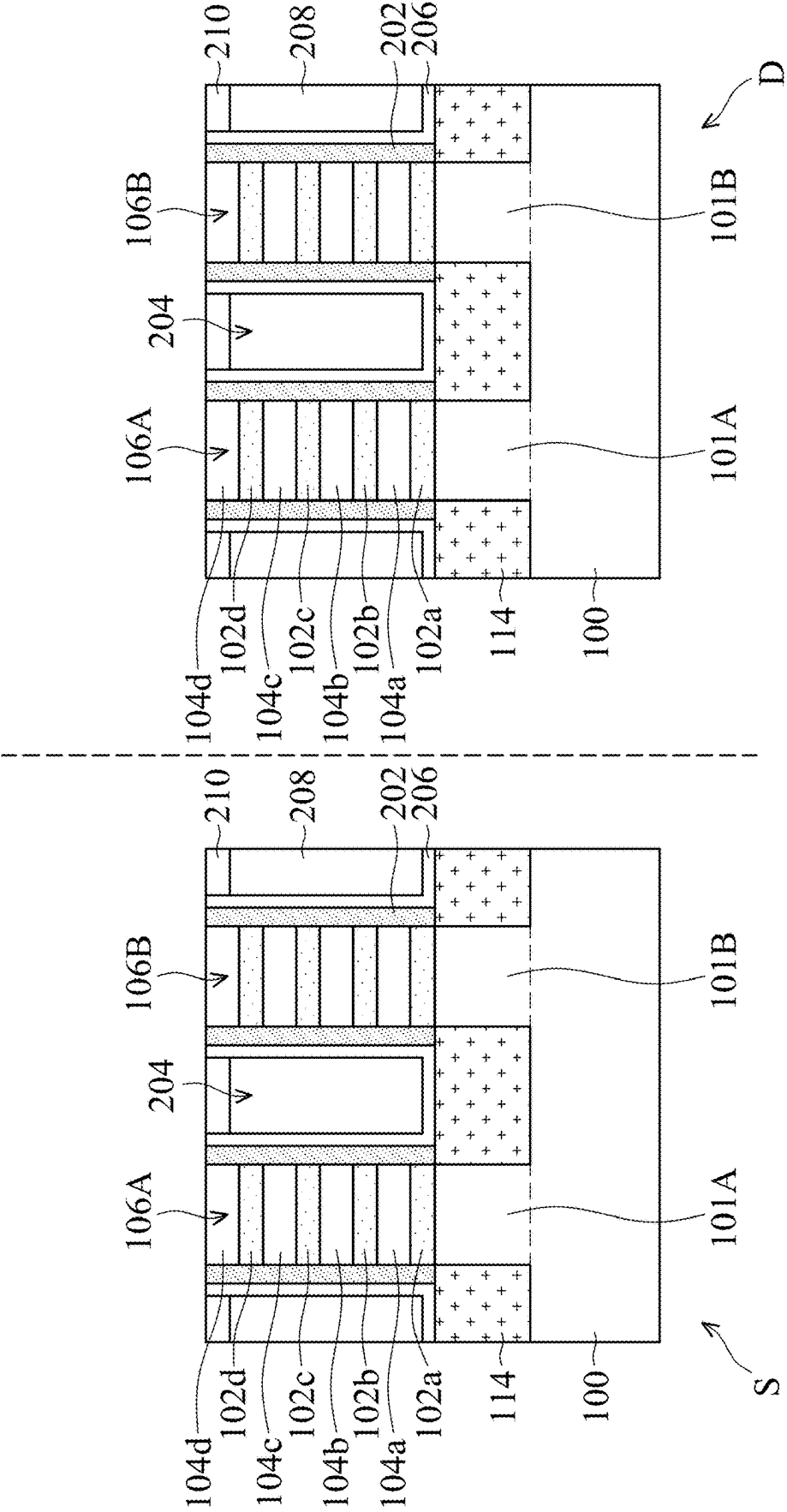


FIG. 4A

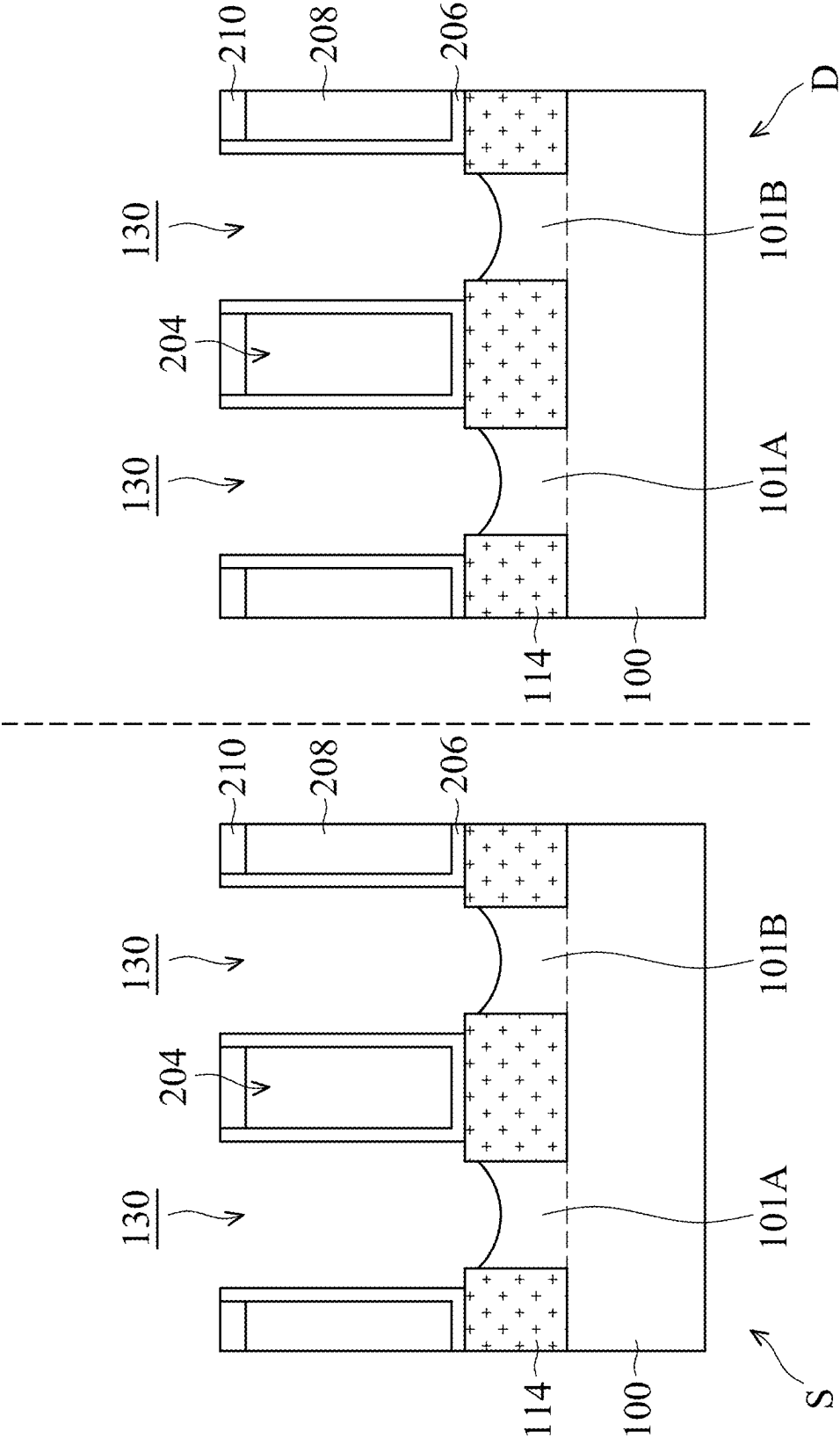


FIG. 4B

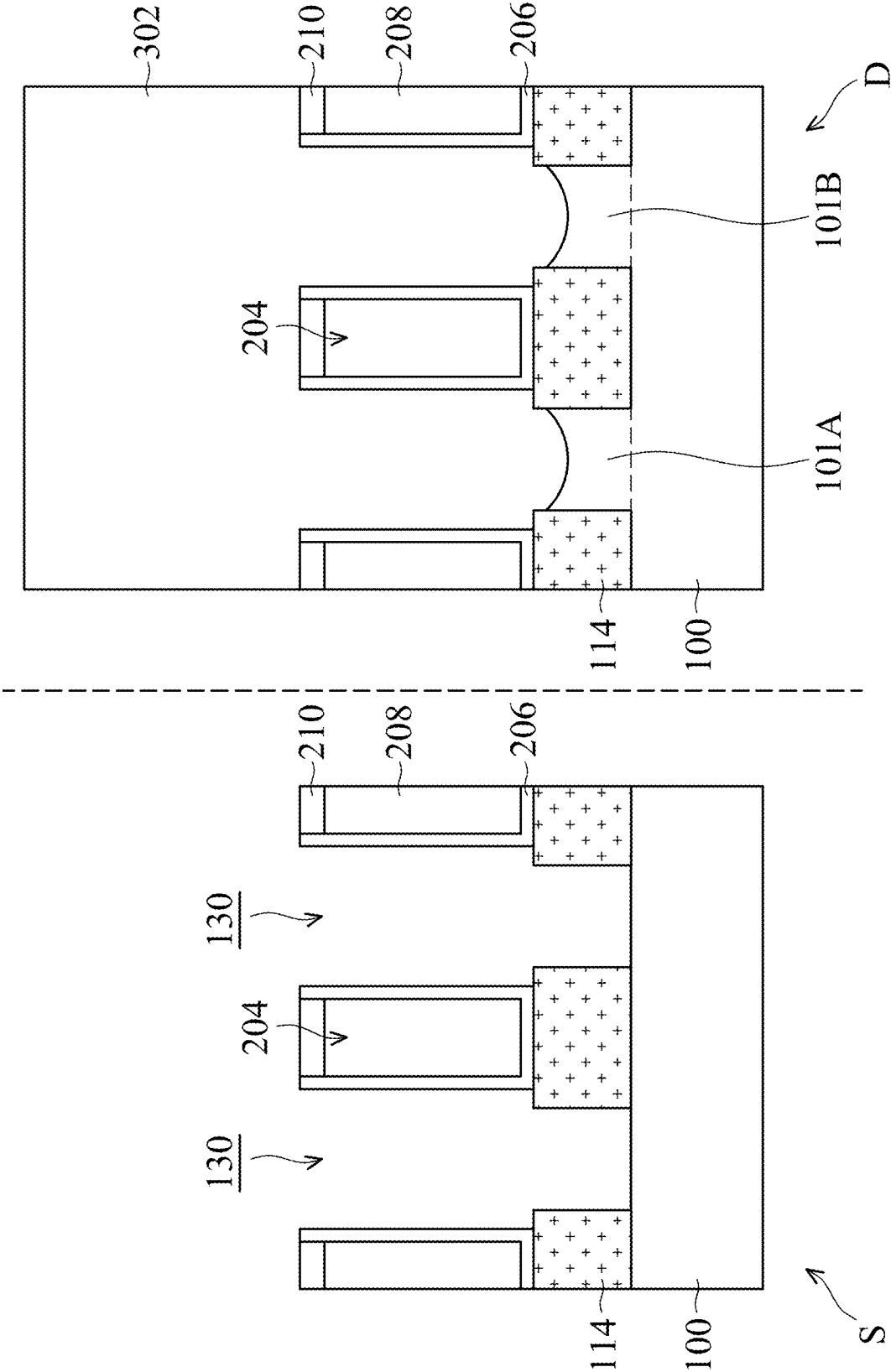


FIG. 4C

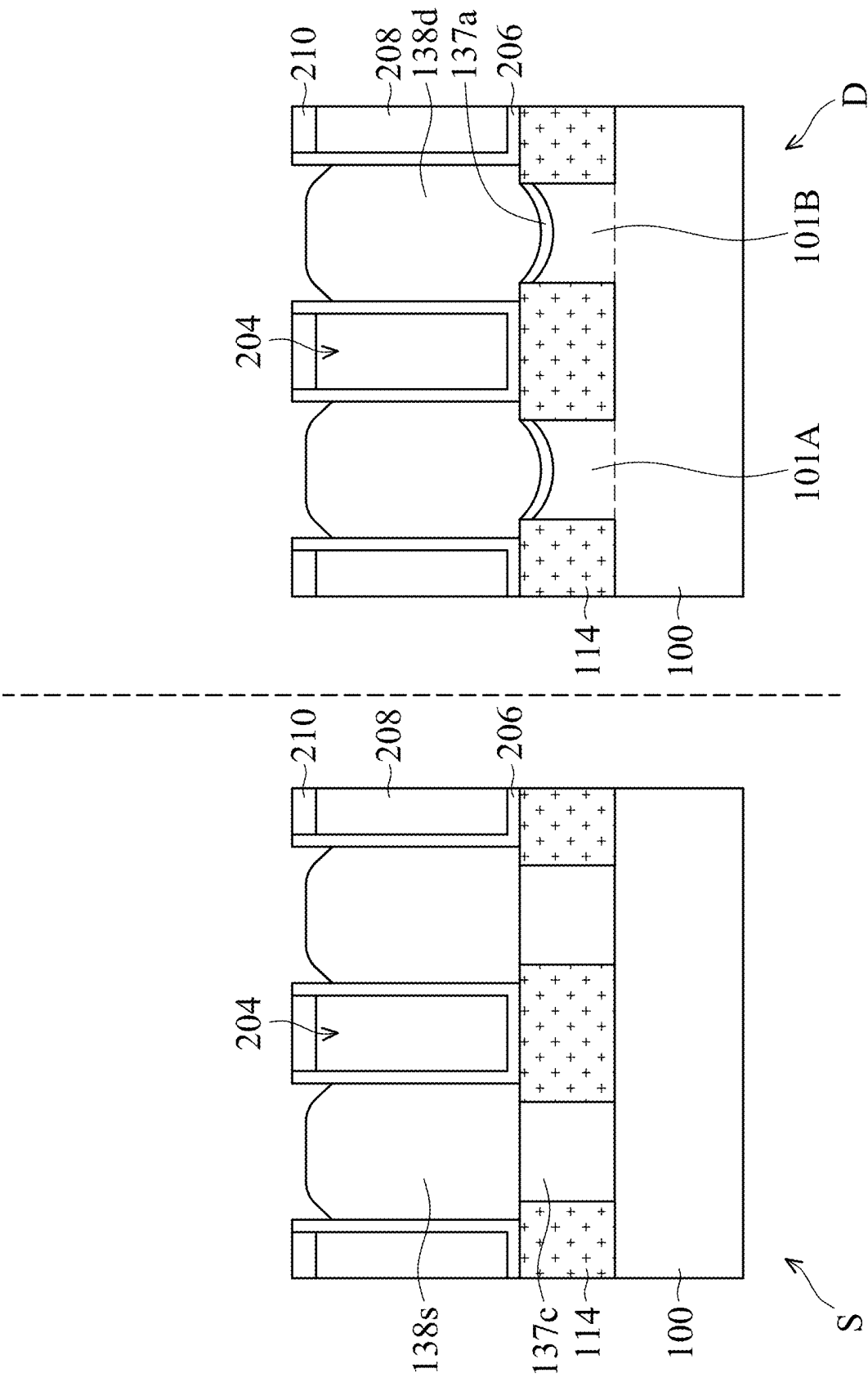


FIG. 4D

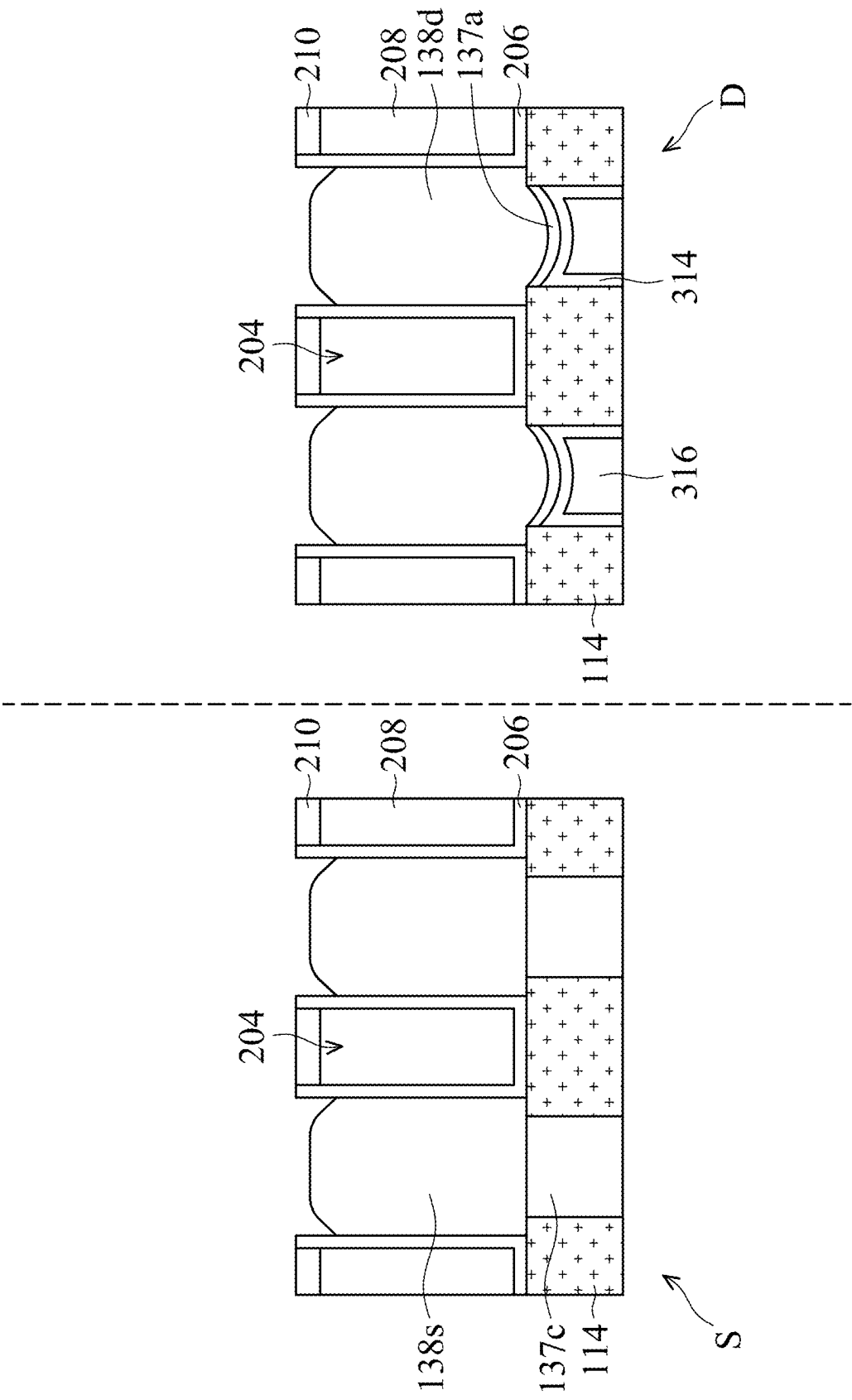


FIG. 4E

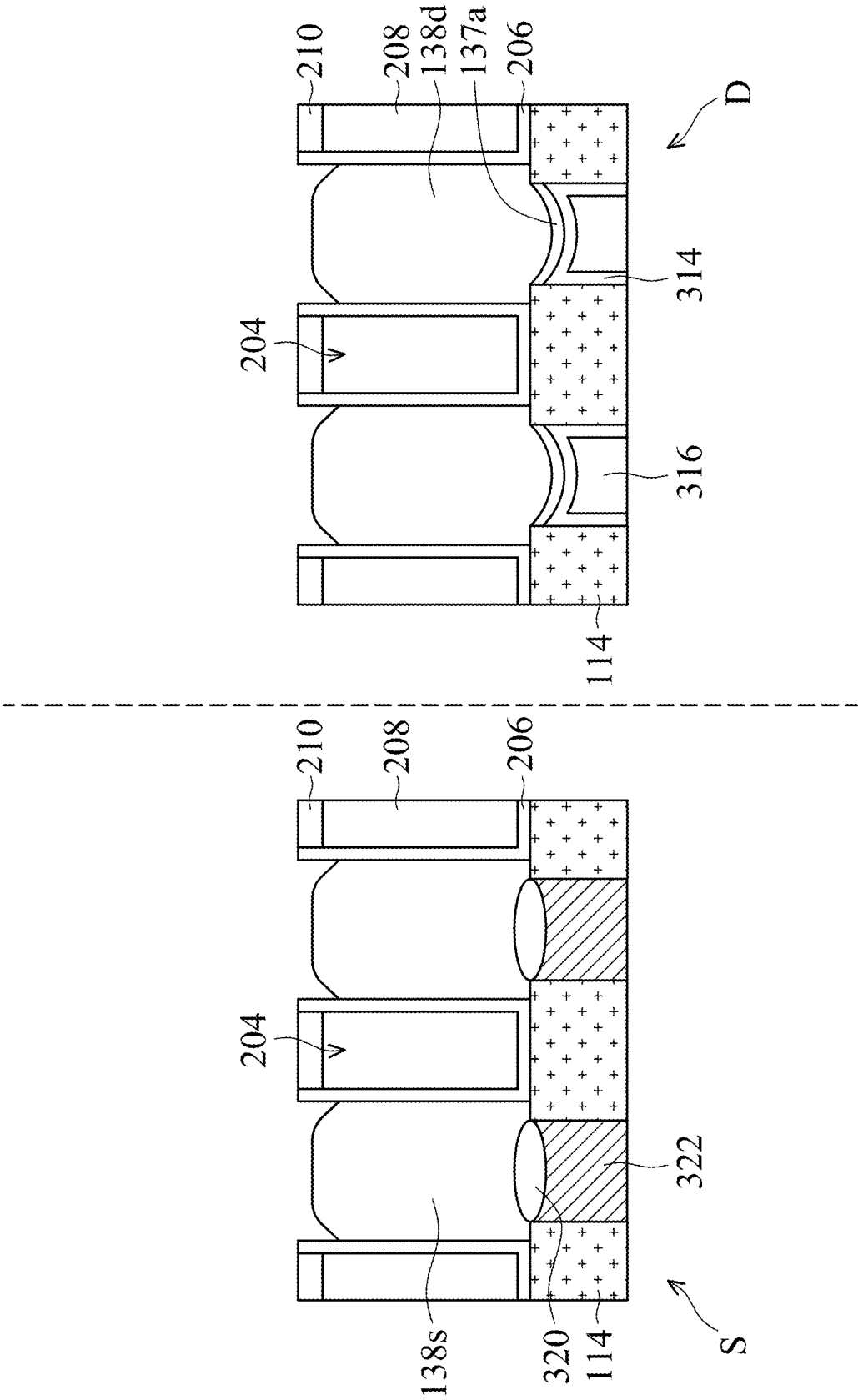


FIG. 4F

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SEMICONDUCTOR DEVICE STRUCTURE WITH BACKSIDE CONTACT

This Application is a Continuation of U.S. application Ser. No. 17/357,052, filed on Jun. 24, 2021, the entirety of which is incorporated by reference herein.

BACKGROUND

The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs. Each generation has smaller and more complex circuits than the previous generation.

Over the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling-down process generally provides benefits by increasing production efficiency and lowering associated costs.

However, these advances have increased the complexity of processing and manufacturing ICs. Since feature sizes continue to decrease, fabrication processes continue to become more difficult to perform. Therefore, it is a challenge to form reliable semiconductor devices at smaller and smaller sizes.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1A-1B are top views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

FIGS. 2A-2J are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

FIGS. 3A-3T are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

FIGS. 4A-4F are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and

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clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

The term “substantially” in the description, such as in “substantially flat” or in “substantially coplanar”, etc., will be understood by the person skilled in the art. In some embodiments the adjective substantially may be removed. Where applicable, the term “substantially” may also include embodiments with “entirely”, “completely”, “all”, etc. Where applicable, the term “substantially” may also relate to 90% or higher, such as 95% or higher, especially 99% or higher, including 100% of what is specified. Furthermore, terms such as “substantially parallel” or “substantially perpendicular” are to be interpreted as not to exclude insignificant deviation from the specified arrangement and may include for example deviations of up to 10 degrees in some embodiments. The word “substantially” does not exclude “completely” e.g. a composition which is “substantially free” from Y may be completely free from Y in some embodiments.

Terms such as “about” in conjunction with a specific distance or size are to be interpreted so as not to exclude insignificant deviation from the specified distance or size and may include for example deviations of up to 10% of what is specified in some embodiments. The term “about” in relation to a numerical value x may mean $x \pm 5$ or 10% of what is specified in some embodiments.

Embodiments of the disclosure may relate to FinFET structure having fins. The fins may be patterned using any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in some embodiments, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins. However, the fins may be formed using one or more other applicable processes.

Embodiments of the disclosure may relate to the gate all around (GAA) transistor structures. The GAA structure may be patterned using any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. In some embodiments, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in some embodiments, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned

sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

Some embodiments of the disclosure are described. Additional operations can be provided before, during, and/or after the stages described in these embodiments. Some of the stages that are described can be replaced or eliminated for different embodiments. Additional features can be added to the semiconductor device structure. Some of the features described below can be replaced or eliminated for different embodiments. Although some embodiments are discussed with operations performed in a particular order, these operations may be performed in another logical order.

FIGS. 2A-2J are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. As shown in FIG. 2A, a semiconductor substrate **100** is received or provided. In some embodiments, the semiconductor substrate **100** is a bulk semiconductor substrate, such as a semiconductor wafer. The semiconductor substrate **100** may include silicon or other elementary semiconductor materials such as germanium. The semiconductor substrate **100** may be un-doped or doped (e.g., p-type, n-type, or a combination thereof). In some embodiments, the semiconductor substrate **100** includes an epitaxially grown semiconductor layer on a dielectric layer. The epitaxially grown semiconductor layer may be made of silicon germanium, silicon, germanium, one or more other suitable materials, or a combination thereof.

In some other embodiments, the semiconductor substrate **100** includes a compound semiconductor. For example, the compound semiconductor includes one or more III-V compound semiconductors having a composition defined by the formula $Al_{X1}Ga_{X2}In_{X3}As_{Y1}P_{Y2}N_{Y3}Sb_{Y4}$, where $X1, X2, X3, Y1, Y2, Y3,$ and $Y4$ represent relative proportions. Each of them is greater than or equal to zero, and added together they equal 1. The compound semiconductor may include silicon carbide, gallium arsenide, indium arsenide, indium phosphide, one or more other suitable compound semiconductors, or a combination thereof. Other suitable substrate including II-VI compound semiconductors may also be used.

In some embodiments, the semiconductor substrate **100** is an active layer of a semiconductor-on-insulator (SOI) substrate. The SOI substrate may be fabricated using a separation by implantation of oxygen (SIMOX) process, a wafer bonding process, another applicable method, or a combination thereof. In some other embodiments, the semiconductor substrate **100** includes a multi-layered structure. For example, the semiconductor substrate **100** includes a silicon-germanium layer formed on a bulk silicon layer.

As shown in FIG. 2A, a semiconductor stack having multiple semiconductor layers is formed over the semiconductor substrate **100**, in accordance with some embodiments. In some embodiments, the semiconductor stack includes multiple semiconductor layers **102a**, **102b**, **102c**, and **102d**, and the semiconductor stack also includes multiple semiconductor layers **104a**, **104b**, **104c**, and **104d**. In some embodiments, the semiconductor layers **102a-102d** and the semiconductor layers **104a-104d** are laid out alternately, as shown in FIG. 2A.

In the present disclosure, the side of the semiconductor substrate **100** where the semiconductor stack is located is referred to as the frontside. The side opposite to the frontside with respect to the semiconductor substrate **100** is referred to as the backside.

In some embodiments, the semiconductor layers **102a-102d** function as sacrificial layers that will be removed in a

subsequent process to release the semiconductor layers **104a-104d**. The semiconductor layers **104a-104d** that are released may function as channel structures of one or more transistors.

In some embodiments, the semiconductor layers **104a-104d** that will be used to form channel structures are made of a material that is different than that of the semiconductor layers **102a-102d**. In some embodiments, the semiconductor layers **104a-104d** are made of or include silicon or silicon germanium. In some embodiments, the sacrificial layers **102a-102d** are made of or include silicon germanium with different atomic concentrations of germanium than that of the semiconductor layers **104a-104d** to achieve different etching selectivity and/or different oxidation rates during subsequent processing. In some embodiments, the semiconductor layer **104a-104d** are substantially free of germanium. In some embodiments, the semiconductor layer **102a-102d** have a greater atomic concentration of germanium than that of the semiconductor layers **104a-104d**.

The present disclosure contemplates that the semiconductor layers **102a-102d** and the semiconductor layers **104a-104d** include any combination of materials (such as semiconductor materials) that can provide desired etching selectivity, desired oxidation rate differences, and/or desired performance characteristics (e.g., materials that maximize current flow).

In some embodiments, the semiconductor layers **102a-102d** and **104a-104d** are formed using multiple epitaxial growth operations. Each of the semiconductor layers **102a-102d** and **104a-104d** may be formed using a selective epitaxial growth (SEG) process, a CVD process (e.g., a vapor-phase epitaxy (VPE) process, a low-pressure chemical vapor deposition (LPCVD) process, and/or an ultra-high vacuum CVD (UHV-CVD) process), a molecular beam epitaxy process, one or more other applicable processes, or a combination thereof. In some embodiments, the semiconductor layers **102a-102d** and **104a-104d** are grown in-situ in the same process chamber. In some embodiments, the growth of the semiconductor layers **102a-102d** and **104a-104d** are alternately and sequentially performed in the same process chamber to complete the formation of the semiconductor stack. In some embodiments, the vacuum of the process chamber is not broken before the epitaxial growth of the semiconductor stack is accomplished.

Afterwards, hard mask elements are formed over the semiconductor stack to assist in a subsequent patterning of the semiconductor stack. One or more photolithography processes and one or more etching processes are used to pattern the semiconductor stack into fin structures **106A** and **106B**, as shown in FIG. 2B in accordance with some embodiments. The fin structures **106A** and **106B** may be patterned by any suitable method. For example, the fin structures **106A** and **106B** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Double-patterning or multi-patterning processes may combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process.

The semiconductor stack is partially removed to form trenches **112**, as shown in FIG. 2B. Each of the fin structures **106A** and **106B** may include portions of the semiconductor layers **102a-102d** and **104a-104d** and semiconductor fins **101A** and **101B**. The semiconductor substrate **100** may also be partially removed during the etching process that forms

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the fin structures **106A** and **106B**. Protruding portions of the semiconductor substrate **100** that remain form the semiconductor fins **101A** and **101B**.

Each of the hard mask elements may include a first mask layer **108** and a second mask layer **110**. The first mask layer **108** and the second mask layer **110** may be made of different materials. In some embodiments, the first mask layer **108** is made of a material that has good adhesion to the semiconductor layer **104d**. The first mask layer **108** may be made of silicon oxide, germanium oxide, silicon germanium oxide, one or more other suitable materials, or a combination thereof. In some embodiments, the second mask layer **110** is made of a material that has good etching selectivity to the semiconductor layers **102a-102d** and **104a-104d**. The second layer **110** may be made of silicon nitride, silicon oxynitride, silicon carbide, one or more other suitable materials, or a combination thereof.

FIGS. 1A-1B are top views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. In some embodiments, the fin structures **106A** and **106B** are oriented lengthwise. In some embodiments, the extending directions of the fin structures **106A** and **106B** are substantially parallel to each other, as shown in FIG. 1A. In some embodiments, FIG. 2B is a cross-sectional view of the structure taken along the line 2B-2B in FIG. 1A.

As shown in FIG. 2C, an isolation structure **114** is formed to surround lower portions of the fin structures **106A** and **106B**, in accordance with some embodiments. In some embodiments, one or more dielectric layers are deposited over the fin structures **106A** and **106B** and the semiconductor substrate **100** to overfill the trenches **112**. The dielectric layers may be made of silicon oxide, silicon oxynitride, borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), low-k material, porous dielectric material, one or more other suitable materials, or a combination thereof. The dielectric layers may be deposited using a flowable chemical vapor deposition (FCVD) process, an atomic layer deposition (ALD) process, a chemical vapor deposition (CVD) process, one or more other applicable processes, or a combination thereof.

Afterwards, a planarization process is used to partially remove the dielectric layers. The hard mask elements (including the first mask layer **108** and the second mask layer **110**) may also function as a stop layer of the planarization process. The planarization process may include a chemical mechanical polishing (CMP) process, a grinding process, a dry polishing process, an etching process, one or more other applicable processes, or a combination thereof. Afterwards, one or more etching back processes are used to partially remove the dielectric layers. As a result, the remaining portion of the dielectric layers forms the isolation structure **114**. Upper portions of the fin structures **106A** and **106B** protrude from the top surface of the isolation structure **114**, as shown in FIG. 2C.

In some embodiments, the etching back process for forming the isolation structure **114** is carefully controlled to ensure that the isolation structure **114** has a suitable height (thickness) H_1 . In some embodiments, the isolation structure **114** is relatively shorter. The height H_1 of the isolation structure **114** may be in a range from about 70 nm to about 90 nm.

Afterwards, the hard mask elements (including the first mask layer **108** and the second mask layer **110**) are removed. Alternatively, in some other embodiments, the hard mask

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elements are removed or consumed during the planarization process and/or the etching back process that forms the isolation structure **114**.

As shown in FIG. 2D, sacrificial spacers **202** are formed on the fin structures **106A** and **106B**, in accordance with some embodiments. In some embodiments, the sacrificial spacers **202** are selectively formed only on the semiconductor materials. In some embodiments, the sacrificial spacers **202** are epitaxially grown on the exposed surfaces of the fin structures **106A** and **106B**. The material and formation method of the sacrificial spacers **202** may be the same as or similar to those of the sacrificial layers **102a-102d**. A suitable epitaxial growth time is used to form the sacrificial spacers **202** to ensure that each of the sacrificial spacers **202** is formed to have a suitable thickness.

As shown in FIG. 2E, dielectric layers **206** and **208** are sequentially deposited, in accordance with some embodiments. Then, a planarization process is used to partially remove the dielectric layers **206** and **208**. In some embodiments, after the planarization process, the top surfaces of the sacrificial spacers **202** and the dielectric layers **206** and **208** are substantially level with each other. The planarization process may include a CMP process, a grinding process, a dry polishing process, an etching process, one or more other applicable processes, or a combination thereof.

The dielectric layer **206** may be made of or include silicon nitride, silicon oxynitride, carbon-containing silicon nitride, carbon-containing silicon oxynitride, one or more other suitable materials, or a combination thereof. The dielectric layer **206** may be deposited using a CVD process, an ALD process, one or more other applicable processes, or a combination thereof. The dielectric layer **208** may be made of or include silicon oxide, silicon carbide, silicon oxycarbide, silicon nitride, silicon oxynitride, carbon-containing silicon nitride, carbon-containing silicon oxynitride, one or more other suitable materials, or a combination thereof. The dielectric layer **208** may be deposited using an FCVD process, a CVD process, an ALD process, one or more other suitable materials, or a combination thereof.

In some embodiments, the dielectric layer **208** is partially removed, in accordance with some embodiments. After the removal of the upper portions of the dielectric layer **208**, multiple recesses are formed. Afterwards, protection structures **210** are formed in the recesses, as shown in FIG. 2F in accordance with some embodiments. As a result, dielectric fins **204** that include the dielectric layers **206** and **208** and the protection structures **210** are formed. The dielectric fins **204** may function as blocking structures that prevent the nearby epitaxial structures (that will be subsequently formed) over nearby fin structures from being merged together. In some embodiments, the dielectric fins **204** are in direct contact with the isolation structure **114**.

In some embodiments, a protection layer is deposited to overfill the recesses. Afterwards, a planarization process is used to remove the portion of the protection layer outside of the recesses. As a result, the remaining portions of the protection layer in the recesses form the protection structures **210**. The planarization process may include a CMP process, a grinding process, a dry polishing process, an etching process, one or more other applicable processes, or a combination thereof.

In some embodiments, the protection layer used for forming the protection structures **210** has a dielectric constant that is different than that of the dielectric layer **206**. In some embodiments, the protection layer used for forming the protection structures **210** has a greater dielectric constant than that of the dielectric layer **206**. The protection struc-

tures **210** may have a dielectric constant that is greater than about 7. The protection layer used for forming the protection structures **210** may be made of or include hafnium oxide, zirconium oxide, aluminum hafnium oxide, aluminum oxide, hafnium silicon oxide, one or more other suitable materials, or a combination thereof. The protection layer used for forming the protection structures **210** may be deposited using a CVD process, an ALD process, one or more other applicable processes, or a combination thereof.

However, embodiments of the disclosure are not limited thereto. Many variations and/or modifications can be made to embodiments of the disclosure. In some other embodiments, the protection layer used for forming the protection structures **210** has a lower dielectric constant than that of the dielectric layer **206**.

Many variations and/or modifications can be made to embodiments of the disclosure. In some other embodiments, the protection structures **210** are not formed.

As shown in FIG. 2F, each of the dielectric fins **204** has a height H_2 . The height H_2 may be in a range from about 70 nm to about 120 nm. The height ratio (H_2/H_1) of the height H_2 to the height H_1 may be in a range from about 0.75 to about 1.7 or 1.8. In some embodiments, the height H_2 of the dielectric fin **204** is greater than the height H_1 of the isolation structure **114**. The height ratio (H_2/H_1) may be in a range from about 1.1 to about 1.8. However, embodiments of the disclosure are not limited thereto. In some other embodiments, the dielectric fin **204** is substantially as high as the isolation structure **114**. In these cases, the height ratio (H_2/H_1) is about 1. In some other embodiments, the dielectric fin **204** is shorter than the isolation structure **114**. The height H_2 is smaller than the height H_1 . In these cases, the height ratio (H_2/H_1) may be in a range from about 0.75 to about 0.99.

Afterwards, dummy gate stacks **120A** and **120B** are formed to extend across the fin structures **106A** and **106B**, as shown in FIG. 1B in accordance with some embodiments. In some embodiments, FIG. 2G is a cross-sectional view of the structure taken along the line 2G-2G in FIG. 1B. FIGS. 3A-3T are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. In some embodiments, FIG. 3A is a cross-sectional view of the structure taken along the line 3A-3A in FIG. 1B.

As shown in FIGS. 1B, 2G, and 3A, the dummy gate stacks **120A** and **120B** are formed to partially cover and to extend across the fin structures **106A** and **106B**, in accordance with some embodiments. In some embodiments, the dummy gate stacks **120A** and **120B** are wrapped around the fin structures **106A** and **106B**. As shown in FIG. 2G, the dummy gate stack **120B** extends across the fin structures **106A** and **106B** and the dielectric fins **204**.

As shown in FIGS. 2G and 3A, each of the dummy gate stacks **120A** and **120B** includes a dummy gate dielectric layer **116** and a dummy gate electrode **118**. The dummy gate dielectric layer **116** may be made of or include silicon oxide. The dummy gate electrodes **118** may be made of or include polysilicon.

In some embodiments, a dummy gate dielectric material layer and a dummy gate electrode layer are sequentially deposited over the isolation structure **114**, the fin structures **106A** and **106B**, and the dielectric fins **204**. The dummy gate dielectric material layer may be deposited using an ALD process, a CVD process, one or more other applicable processes, or a combination thereof. The dummy gate electrode layer may be deposited using a CVD process. Afterwards, the dummy gate dielectric material layer and the

dummy gate electrode layer are patterned to form the dummy gate stacks **120A** and **120B**.

In some embodiments, hard mask elements including mask layers **122** and **124** are used to assist in the patterning process for forming the dummy gate stacks **120A** and **120B**. With the hard mask elements as an etching mask, one or more etching processes are used to partially remove the dummy gate dielectric material layer and the dummy gate electrode layer. As a result, remaining portions of the dummy gate dielectric material layer and the dummy gate electrode layer form the dummy gate stacks **120A** and **120B** that include the dummy gate dielectric layer **116** and the dummy gate electrodes **118**.

FIGS. 4A-4F are cross-sectional views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. In some embodiments, FIG. 4A is a cross-sectional view of the structure taken along the lines 4A-4A and 4A'-4A' in FIG. 1B. As shown in FIG. 4A, the left side shows a first region S that is the cross-sectional view of the structure taken along the line 4A-4A, and the right side shows a second region D that is the cross-sectional view of the structure taken along the line 4A'-4A'. In some embodiments, the first region S shows the elements near a subsequently formed source region, and the second region D shows the elements near a subsequently formed drain region.

As shown in FIG. 3B, spacer layers **126** and **128** are afterwards deposited over the structure shown in FIG. 3A, in accordance with some embodiments. The spacer layers **126** and **128** extend along the sidewalls of the dummy gate stacks **120A** and **120B**. The spacer layers **126** and **128** are made of different materials. The spacer layer **126** may be made of a dielectric material that has a low dielectric constant. The spacer layer **126** may be made of or include silicon carbide, silicon oxycarbide, silicon oxide, one or more other suitable materials, or a combination thereof. The spacer layer **128** may be made of a dielectric material that can provide more protection to the gate stacks during subsequent processes. The spacer layer **128** may have a greater dielectric constant than that of the spacer layer **126**. The spacer layer **128** may be made of silicon nitride, silicon oxynitride, carbon-containing silicon nitride, carbon-containing silicon oxynitride, one or more other suitable materials, or a combination thereof. The spacer layers **126** and **128** may be sequentially deposited using a CVD process, an ALD process, a physical vapor deposition (PVD) process, one or more other applicable processes, or a combination thereof.

As shown in FIG. 3C, the spacer layers **126** and **128** are partially removed, in accordance with some embodiments. One or more anisotropic etching processes may be used to partially remove the spacer layers **126** and **128**. As a result, remaining portions of the spacer layers **126** and **128** form spacer elements **126'** and **128'**, respectively. The spacer elements **126'** and **128'** extend along the sidewalls of the dummy gate stacks **120A** and **120B**, as shown in FIG. 3C.

As shown in FIGS. 3C and 4B, the fin structures **106A** and **106B** are partially removed to form recesses **130**, in accordance with some embodiments. The recesses **130** may be used to contain epitaxial structures (such as source/drain structures) that will be formed later. As shown in FIG. 4B, the recesses **130** in the first region S are used to contain source structures that will be formed later. The recesses **130** in the second region D are used to contain drain structures that will be formed later.

One or more etching processes may be used to form the recesses **130**. In some embodiments, a dry etching process is used to form the recesses **130**. Alternatively, a wet etching

process may be used to form the recesses 130. In some embodiments, each of the recesses 130 penetrates into the fin structure 106A or 106B. In some embodiments, the recesses 130 further extend into the semiconductor fin 101A or 101B, as shown in FIGS. 3C and 4B. In some embodiments, the spacer elements 126' and 128' and the recesses 130 are simultaneously formed using the same etching process.

In some embodiments, each of the recesses 130 has slanted sidewalls, as shown in FIG. 3C. Upper portions of the recesses 130 are larger (or wider) than lower portions of the recesses 130. In these cases, due to the profile of the recesses 130, an upper semiconductor layer (such as the semiconductor layer 104d) is shorter than a lower semiconductor layer (such as the semiconductor layer 104b).

However, embodiments of the disclosure have many variations. In some other embodiments, the recesses 130 have substantially vertical sidewalls. In these cases, due to the profile of the recesses 130, an upper semiconductor layer (such as the semiconductor layer 104d) is substantially as wide as a lower semiconductor layer (such as the semiconductor layer 104b).

As shown in FIG. 3D, the semiconductor layers 102a-102d are laterally etched, in accordance with some embodiments. As a result, edges of the semiconductor layers 102a-102d retreat from edges of the semiconductor layers 104a-104d. As shown in FIG. 3D, recesses 132 are formed due to the lateral etching of the semiconductor layers 102a-102d. The recesses 132 may be used to contain inner spacers that will be formed later. The semiconductor layers 102a-102d may be laterally etched using a wet etching process, a dry etching process, or a combination thereof. In some other embodiments, the semiconductor layers 102a-102d are partially oxidized before being laterally etched.

During the lateral etching of the semiconductor layers 102a-102d, the semiconductor layers 104a-104d may also be slightly etched. As a result, edge portions of the semiconductor layers 104a-104d are partially etched and thus shrink to become edge elements 105a-105d, as shown in FIG. 3D. As shown in FIG. 3D, each of the edge elements 105a-105d of the semiconductor layers 104a-104d is thinner than the corresponding inner portion of the semiconductor layers 104a-104d.

As shown in FIG. 3E, an insulating layer 134 is deposited over the structure shown in FIG. 3D, in accordance with some embodiments. The insulating layer 134 covers the dummy gate stacks 120A and 120B and fills the recesses 132. The insulating layer 134 may be made of or include carbon-containing silicon nitride (SiCN), carbon-containing silicon oxynitride (SiOCN), carbon-containing silicon oxide (SiOC), silicon oxide, silicon nitride, one or more other suitable materials, or a combination thereof. In some embodiments, the insulating layer 134 is a single layer. In some other embodiments, the insulating layer 134 includes multiple sub-layers. Some of the sub-layers may be made of different materials and/or contain different compositions. The insulating layer 134 may be deposited using a CVD process, an ALD process, one or more other applicable processes, or a combination thereof.

As shown in FIG. 3F, an etching process is used to partially remove the insulating layer 134, in accordance with some embodiments. The remaining portions of the insulating layer 134 form inner spacers 136, as shown in FIG. 3F. The etching process may include a dry etching process, a wet etching process, or a combination thereof.

The inner spacers 136 cover the edges of the semiconductor layers 102a-102d. The inner spacers 136 may be used to prevent subsequently formed epitaxial structures (that

function as, for example, source/drain structures) from being damaged during a subsequent process for removing the sacrificial layers 102a-102d. In some embodiments, the inner spacers 136 are made of a low-k material that has a lower dielectric constant than that of silicon oxide. In these cases, the inner spacers 136 may also be used to reduce parasitic capacitance between the subsequently formed source/drain structures and the gate stacks. As a result, the operation speed of the semiconductor device structure may be improved.

In some embodiments, after the etching process for forming the inner spacers 136, portions of the semiconductor fin 101A originally covered by the insulating layer 134 are exposed by the recesses 130, as shown in FIG. 3F. The edges of the semiconductor layers 104a-104d are exposed by the recesses 130, as shown in FIG. 3F.

As shown in FIGS. 3G and 4C, a mask element 302 is formed to cover the recesses 130 that are designed to contain the subsequently formed drain structures, in accordance with some embodiments. As shown in FIGS. 3G, and 4C, the mask element 302 has an opening 304 that exposes the recesses 130 that are designed to contain the subsequently formed source structures. Afterwards, one or more etching processes are used to partially remove the semiconductor fins 101A and 101B. As a result, the recesses 130 further extend towards the semiconductor substrate 100. In some embodiments, the extended recesses 130 expose the semiconductor substrate 100, as shown in FIGS. 3G and 4C.

As shown in FIG. 3H, epitaxial structures 138s and 138d are formed beside the dummy gate stacks 120A and 120B, in accordance with some embodiments. In some embodiments, the epitaxial structures 138s and 138d fill the recesses 130, as shown in FIGS. 3H and 4D. In some other embodiments, the epitaxial structures 138s and 138d overfill the recesses 130. In these cases, the top surfaces of the epitaxial structures 138s and 138d may be higher than the top surface of the dummy gate dielectric layer 116. In some other embodiments, the epitaxial structures 138s and 138d partially fill the recesses 130. As shown in FIG. 4D, the dielectric fins 204 may be used to prevent the nearby epitaxial structures 138s or the nearby epitaxial structures 138d from being merged together. In some embodiments, the epitaxial structures 138s are in direct contact with the dielectric fins 204, as shown in FIG. 4D. In some embodiments, the height of each of the epitaxial structures 138s is greater than the height of the isolation structure 114.

In some embodiments, the epitaxial structures 138s and 138d connect to the semiconductor layers 104a-104d. Each of the semiconductor layers 104a-104d is sandwiched between the epitaxial structures 138s and 138d. In some embodiments, the epitaxial structures 138s function as source structures, and the epitaxial structures 138d function as drain structures. In some embodiments, some of the epitaxial structures 138s and 138d are p-type doped regions. The epitaxial structures 138s and 138d may include epitaxially grown silicon germanium (SiGe), epitaxially grown silicon, or another suitable epitaxially grown semiconductor material.

However, embodiments of the disclosure are not limited thereto. In some other embodiments, some of the epitaxial structures 138s and 138d are n-type doped regions. The epitaxial structures 138s and 138d may include epitaxially grown silicon, epitaxially grown silicon carbide (SiC), epitaxially grown germanium, or another suitable epitaxially grown semiconductor material.

In some embodiments, each of the epitaxial structures 138d includes a first portion 137a and second portions 137b.

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The first portion **137a** is positioned at the bottom portion of the epitaxial structure **138d**. Each of the epitaxial structures **138s** includes a bottom portion **137c** and second portions **137b**. The bottom portion **137c** is positioned at the bottom portion of the epitaxial structure **138s**. The bottom portion **137c** may fill the extended portion of the recess **130**. The bottom portion **137c** may be surrounded by the isolation structure **114**, as shown in FIG. 4D. The bottom portion **137c** may function as a dummy epitaxial structure and will be removed later.

The second portions **137b** are positioned adjacent to the semiconductor layer **104b**, **104c**, or **104d**. The first portion **137a** and the second portions **137b** may have different compositions than other portions of the epitaxial structure **138s** or **138d**. In some embodiments, the first portion **137a**, the second portions **137b**, and the bottom portion **137c** have a lower dopant concentration than the other portions of the epitaxial structure **138s** or **138d**. In some embodiments, the first portion **137a** or the bottom portion **137c** is substantially free of dopant. The first portion **137a** may be used to prevent or reduce current leakage from the epitaxial structures **138s** and **138d**. In some embodiments, the first portion **137a** and the second portions **137b** have a lower atomic concentration of germanium than the other portions of the epitaxial structure **138s** or **138d**.

In some embodiments, the epitaxial structures **138s** and **138d** are formed using a selective epitaxial growth (SEG) process, a CVD process (e.g., a vapor-phase epitaxy (VPE) process, a low-pressure chemical vapor deposition (LPCVD) process, and/or an ultra-high vacuum CVD (UHV-CVD) process), a molecular beam epitaxy process, one or more other applicable processes, or a combination thereof. In some embodiments, the formation of the epitaxial structures **138s** and **138d** involve one or more etching processes.

In some embodiments, the epitaxial structures **138s** and **138d** are doped with one or more suitable p-type dopants. For example, the epitaxial structures **138s** and **138d** are SiGe source/drain features or Si source/drain features that are doped with boron (B), gallium (Ga), indium (In), or another suitable dopant. In some other embodiments, the epitaxial structures **138s** and **138d** are doped with one or more suitable n-type dopants. For example, the epitaxial structures **138s** and **138d** are Si source/drain features doped with phosphor (P), antimony (Sb), or another suitable dopant.

In some embodiments, the epitaxial structures **138s** and **138d** are doped in-situ during their epitaxial growth. The initial reaction gas mixture for forming the epitaxial structures **138s** and **138d** contains dopants. In some other embodiments, the epitaxial structures **138s** and **138d** are not doped during the growth of the epitaxial structures **138s** and **138d**. Instead, after the formation of the epitaxial structures **138s** and **138d**, the epitaxial structures **138s** and **138d** are doped in a subsequent process. In some embodiments, the doping is achieved by using an ion implantation process, a plasma immersion ion implantation process, a gas and/or solid source diffusion process, one or more other applicable processes, or a combination thereof. In some embodiments, the epitaxial structures **138s** and **138d** are further exposed to one or more annealing processes to activate the dopants. For example, a rapid thermal annealing process is used.

As shown in FIG. 3I, a contact etch stop layer **139** and a dielectric layer **140** are formed to cover the epitaxial structures **138s** and **138d**, and to surround the dummy gate stacks **120A** and **120B**, in accordance with some embodiments. The contact etch stop layer **139** may be made of or include silicon nitride, silicon oxynitride, silicon carbide, aluminum

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oxide, one or more other suitable materials, or a combination thereof. The dielectric layer **140** may be made of or include silicon oxide, silicon oxynitride, borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), low-k material, porous dielectric material, one or more other suitable materials, or a combination thereof.

In some embodiments, an etch stop material layer and a dielectric material layer are sequentially deposited over the structure shown in FIG. 3H. The etch stop material layer may be deposited using a CVD process, an ALD process, a PVD process, one or more other applicable processes, or a combination thereof. The dielectric material layer may be deposited using an FCVD process, a CVD process, an ALD process, one or more other applicable processes, or a combination thereof.

Afterwards, a planarization process is used to partially remove the etch stop material layer and the dielectric material layer. As a result, the remaining portions of the etch stop material layer and the dielectric material layer form the contact etch stop layer **139** and the dielectric layer **140**, respectively. The planarization process may include a CMP process, a grinding process, an etching process, a dry polishing process, one or more other applicable processes, or a combination thereof. In some embodiments, the mask layers **122** and **124** are removed during the planarization process. In some embodiments, after the planarization process, the top surfaces of the contact etch stop layer **139**, the dielectric layer **140**, and the dummy gate electrodes **118** are substantially coplanar.

As shown in FIGS. 2H and 3J, one or more etching processes are used to remove the dummy gate electrodes **118**, in accordance with some embodiments. As a result, trenches **142** are formed. The trenches **142** are surrounded by the dielectric layer **140**. The trenches **142** expose the dummy gate dielectric layer **116**.

As shown in FIGS. 2I and 3K, the dummy gate dielectric layer **116**, the semiconductor layers **102a-102d** (that function as sacrificial layers), and the sacrificial spacers **202** are removed, in accordance with some embodiments. In some embodiments, an etching process is used to remove the semiconductor layers **102a-102d** and the sacrificial spacers **202**. As a result, recesses **144** are formed, as shown in FIGS. 2I and 3K.

Due to high etching selectivity, the semiconductor layers **104a-104d** are slightly (or substantially not) etched. The remaining portions of the semiconductor layers **104a-104d** form multiple semiconductor nanostructures **104a'-104d'** of the fin structures **106A** and **106B**, as shown in FIGS. 2I and 3K. The semiconductor nanostructures **104a'-104d'** are constructed by or made up of the remaining portions of the semiconductor layers **104a-104d**. The semiconductor nanostructures **104a'-104d'** suspended over the semiconductor fin **101A** or **101B** may function as channel structures of transistors.

In some embodiments, the etchant used for removing the semiconductor layers **102a-102d** also slightly removes the semiconductor layers **104a-104d** that form the semiconductor nanostructures **104a'-104d'**. As a result, the obtained semiconductor nanostructures **104a'-104d'** become thinner after the removal of the semiconductor layers **102a-102d**. In some embodiments, each of the semiconductor nanostructures **104a'-104d'** is thinner than each of the edge portions **105b-105d** since the edge portions **105b-105d** are surrounded by other elements and thus are prevented from being reached and etched by the etchant.

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However, embodiments of the disclosure are not limited thereto. In some other embodiments, each of the semiconductor nanostructures **104a'-104d'** is substantially as thick as each of the edge portions **105b-105d**. In some other embodiments, each of the semiconductor nanostructures **104a'-104d'** is thicker than each of the edge portions **105b-105d**.

After the removal of the semiconductor layers **102a-102d** (that function as sacrificial layers), the recesses **144** are formed. The recesses **144** connect to the trench **142** and surround each of the semiconductor nanostructures **104a'-104d'**. As shown in FIGS. 2I and 3K, even if the recesses **144** between the semiconductor nanostructures **104a'-104d'** are formed, the semiconductor nanostructures **104a'-104d'** remain held by the epitaxial structures **138s** and **138d** (see FIG. 3K). Therefore, after the removal of the semiconductor layers **102a-102d** (that function as sacrificial layers), the released semiconductor nanostructures **104a'-104d'** are prevented from falling down.

During the removal of the semiconductor layers **102a-102d** (that function as sacrificial layers), the inner spacers **136** protect the epitaxial structures **138s** and **138d** from being etched or damaged. The quality and reliability of the semiconductor device structure are improved.

As shown in FIGS. 2J and 3L, metal gate stacks **156A** and **156B** are formed to fill the trenches **142**, in accordance with some embodiments. The metal gate stacks **156A** and **156B** extend into the recesses **144** to wrap around each of the semiconductor nanostructures **104a'-104d'**.

Each of the metal gate stacks **156A** and **156B** includes multiple metal gate stack layers. Each of the metal gate stacks **156A** and **156B** may include a gate dielectric layer **150** and a metal gate electrode **152**. The metal gate electrode **152** may include a work function layer. The metal gate electrode **152** may further include a conductive filling. In some embodiments, the formation of the metal gate stacks **156A** and **156B** involves the deposition of multiple metal gate stack layers over the dielectric layer **140** to fill the trenches **142** and the recesses **144**. The metal gate stack layers extend into the recesses **144** to wrap around each of the semiconductor nanostructures **104a'-104d'**.

In some embodiments, the gate dielectric layer **150** is made of or includes a dielectric material with high dielectric constant (high-K). The gate dielectric layer **150** may be made of or include hafnium oxide, zirconium oxide, aluminum oxide, hafnium dioxide-alumina alloy, hafnium silicon oxide, hafnium silicon oxynitride, hafnium tantalum oxide, hafnium titanium oxide, hafnium zirconium oxide, one or more other suitable high-K materials, or a combination thereof. The gate dielectric layer **150** may be deposited using an ALD process, a CVD process, one or more other applicable processes, or a combination thereof.

In some embodiments, before the formation of the gate dielectric layer **150**, interfacial layers are formed on the surfaces of the semiconductor nanostructures **104a'-104d'**. The interfacial layers are very thin and are made of, for example, silicon oxide or germanium oxide. In some embodiments, the interfacial layers are formed by applying an oxidizing agent on the surfaces of the semiconductor nanostructures **104a'-104d'**. For example, a hydrogen peroxide-containing liquid may be applied or provided on the surfaces of the semiconductor nanostructures **104a'-104d'** so as to form the interfacial layers.

The work function layer of the metal gate electrode **152** may be used to provide the desired work function for transistors to enhance device performance including improved threshold voltage. In some embodiments, the work function layer is used for forming a PMOS device. The work

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function layer is a p-type work function layer. The p-type work function layer is capable of providing a work function value suitable for the device, such as equal to or greater than about 4.8 eV.

The p-type work function layer may include metal, metal carbide, metal nitride, other suitable materials, or a combination thereof. For example, the p-type metal includes tantalum nitride, tungsten nitride, titanium, titanium nitride, one or more other suitable materials, or a combination thereof.

In some other embodiments, the work function layer is used for forming an NMOS device. The work function layer is an n-type work function layer. The n-type work function layer is capable of providing a work function value suitable for the device, such as equal to or less than about 4.5 eV.

The n-type work function layer may include metal, metal carbide, metal nitride, or a combination thereof. For example, the n-type work function layer includes titanium nitride, tantalum, tantalum nitride, one or more other suitable materials, or a combination thereof. In some embodiments, the n-type work function is an aluminum-containing layer. The aluminum-containing layer may be made of or include TiAlC, TiAlO, TiAlN, one or more other suitable materials, or a combination thereof.

The work function layer may also be made of or include hafnium, zirconium, titanium, tantalum, aluminum, metal carbides (e.g., hafnium carbide, zirconium carbide, titanium carbide, aluminum carbide), aluminides, ruthenium, palladium, platinum, cobalt, nickel, conductive metal oxides, or a combinations thereof. The thickness and/or the compositions of the work function layer may be fine-tuned to adjust the work function level.

The work function layer may be deposited over the gate dielectric layer **150** using an ALD process, a CVD process, a PVD process, an electroplating process, an electroless plating process, one or more other applicable processes, or a combination thereof.

In some embodiments, a barrier layer is formed before the work function layer to interface the gate dielectric layer **150** with the subsequently formed work function layer. The barrier layer may also be used to prevent diffusion between the gate dielectric layer **150** and the subsequently formed work function layer. The barrier layer may be made of or include a metal-containing material. The metal-containing material may include titanium nitride, tantalum nitride, one or more other suitable materials, or a combination thereof. The barrier layer may be deposited using an ALD process, a CVD process, a PVD process, an electroplating process, an electroless plating process, one or more other applicable processes, or a combination thereof.

In some embodiments, the conductive fillings of the metal gate electrodes **152** are made of or include a metal material. The metal material may include tungsten, aluminum, copper, cobalt, one or more other suitable materials, or a combination thereof. A conductive layer used for forming the conductive filling may be deposited over the work function layer using a CVD process, an ALD process, a PVD process, an electroplating process, an electroless plating process, a spin coating process, one or more other applicable processes, or a combination thereof.

In some embodiments, a blocking layer is formed over the work function layer before the formation of the conductive layer used for forming the conductive filling. The blocking layer may be used to prevent the subsequently formed conductive layer from diffusing or penetrating into the work function layer. The blocking layer may be made of or include tantalum nitride, titanium nitride, one or more other suitable

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materials, or a combination thereof. The blocking layer may be deposited using an ALD process, a PVD process, an electroplating process, an electroless plating process, one or more other applicable processes, or a combination thereof.

Afterwards, a planarization process is performed to remove the portions of the metal gate stack layers outside of the trenches **142**, in accordance with some embodiments. As a result, the remaining portions of the metal gate stack layers form the metal gate stacks **156A** and **156B**, as shown in FIGS. **2J** and **3L**. A portion of the metal gate stack **156B** is between the semiconductor nanostructures **104a'-104d'** and the dielectric fins **204**, as shown in FIG. **2J**.

In some embodiments, the conductive filling does not extend into the recesses **144** since the recesses **144** are small and have been filled with other elements such as the gate dielectric layer **150** and the work function layer. However, embodiments of the disclosure are not limited thereto. In some other embodiments, a portion of the conductive filling extends into the recesses **144**, especially for the lower recesses **144** that may have larger space.

As shown in FIG. **3M**, frontside conductive contacts **158** and an interconnection structure **160** are formed, in accordance with some embodiments. In some embodiments, each of the frontside conductive contacts is electrically connected to one of the epitaxial structures **138d**. In some embodiments, no frontside conductive contact is formed to be in electrical contact with the epitaxial structure **138s**. A backside conductive contact will be formed later to be in electrical contact with the epitaxial structure **138s**. Therefore, the metal layer routing is significantly relaxed.

The formation of the frontside conductive contacts **158** may involve one or more patterning processes for forming contact openings, one or more deposition processes for forming conductive materials into the conductive openings, and one or more planarization processes to accomplish the formation of the frontside conductive contacts **158**. The frontside conductive contacts **158** may be made of or include Co, Ru, W, Ti, TiN, Ta, TaN, Mo, Ni, Cu, Au, one or more other suitable materials, or a combination thereof.

In some embodiments, a metal-semiconductor compound layer (not shown) is formed between the frontside conductive contact **158** and the epitaxial structure **138d**. The metal-semiconductor compound layer may be a metal silicide material that contains titanium, nickel, cobalt, tantalum, platinum, tungsten, one or more other noble metals, one or more other refractory metals, one or more other rare earth metals, one or more other suitable materials, or a combination thereof. Alternatively, the metal-semiconductor compound layer may be a metal-containing material that contains no silicon. For example, the metal-semiconductor compound layer may contain germanium and a metal element such as titanium, nickel, cobalt, tantalum, platinum, tungsten, one or more other noble metals, one or more other refractory metals, one or more other rare earth metals, one or more other suitable materials, or a combination thereof.

The interconnection structure **160** may include middle-end-of line structures and back-end-of-line structures. The interconnection structure **160** may include multiple dielectric layers, multiple conductive contacts, multiple conductive lines, and multiple conductive vias. The interconnection structure **160** may include four to five metal lines that are vertically stacked. The formation of the interconnection structure **160** may involve multiple deposition processes, multiple patterning processes, and multiple planarization processes.

As shown in FIG. **3N**, a carrier substrate **170** is bonded onto the interconnection structure **160**, in accordance with

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some embodiments. The carrier substrate **170** may be a semiconductor wafer such as a silicon wafer, a silicon-on-insulator (SOI) wafer, or a glass wafer. An adhesion layer **162** may be formed to achieve the bonding between the carrier substrate **170** and the interconnection structure **160**. After bonding with the carrier substrate **170**, the structure is flipped upside down such that the backside surface of the semiconductor substrate **100** faces upwards, as shown in FIG. **3N**.

As shown in FIG. **3O**, a thinning process is then used to thin down the semiconductor substrate **100**, in accordance with some embodiments. In some embodiments, after the thinning process, the semiconductor substrate **100** is removed such that the semiconductor fin **101A** and the isolation structure **114** are exposed. In some embodiments, the bottom portion **137c** (that functions as a dummy epitaxial structure) is also exposed, as shown in FIG. **3O**. The thinning process may include a CMP process, a grinding process, a dry polishing process, an etching process, one or more other applicable processes, or a combination thereof.

As shown in FIG. **3P**, with the bottom portion **137c** (that functions as a dummy epitaxial structure) as an etching mask, one or more etching processes are used to partially remove the semiconductor fins (such as the semiconductor fin **101A**), in accordance with some embodiments. As a result, openings **312** that expose the epitaxial structures **138d** are formed.

As shown in FIGS. **3Q** and **4E**, etch stop elements **314** and dielectric fillings **316** are formed in the openings **312**, in accordance with some embodiments. The etch stop elements **314** extend along sidewalls of the bottom portion **137c** (that functions as a dummy epitaxial structure), sidewalls of the isolation structures **114** in the second region D, and bottoms of the epitaxial structures **138d**. The dielectric fillings **316** may fill the remaining space of the openings **312**. In some embodiments, because the isolation structure **114** is formed to have a low height, each of the openings **312** has a low aspect ratio, which facilitates the formation of the etch stop elements **314** and the dielectric fillings **316** in the openings **312**. For example, voids are prevented from being formed in the dielectric fillings **316**. The reliability and quality of the dielectric fillings **316** are improved.

Each of the dielectric fillings **316** may be a single layer structure. Alternatively, each of the dielectric fillings **316** may be a multilayer structure that includes multiple sub-layers. Some of the sub-layers may be made of different materials. The etch stop elements **314** and the dielectric fillings **316** may be used to significantly prevent short circuiting between the backside conductive contact (that will be formed later) and other conductive elements such as a nearby backside conductive contact or a nearby drain structure.

The etch stop elements **314** may be made of or include silicon nitride, silicon oxynitride, silicon carbide, aluminum oxide, one or more other suitable materials, or a combination thereof. The dielectric fillings **316** may be made of or include silicon oxide, silicon oxynitride, borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), low-k material, porous dielectric material, one or more other suitable materials, or a combination thereof.

In some embodiments, an etch stop material layer and a dielectric material layer are sequentially deposited to overfill the openings **312**. The etch stop material layer may be deposited using a CVD process, an ALD process, a PVD process, one or more other applicable processes, or a combination thereof. The dielectric material layer may be depos-

ited using an FCVD process, a CVD process, an ALD process, one or more other applicable processes, or a combination thereof.

Afterwards, a planarization process is used to remove the etch stop material layer and the dielectric material layer outside of the openings 312. As a result, the remaining portions of the etch stop material layer and the dielectric material layer form the etch stop elements 314 and the dielectric fillings 316, respectively. The planarization process may include a CMP process, a grinding process, an etching process, a dry polishing process, one or more other applicable processes, or a combination thereof. In some embodiments, after the planarization process, the surfaces of the etch stop elements 314, the dielectric fillings 316, and the bottom portion 137c (that functions as a dummy epitaxial structure) are substantially level with each other.

As shown in FIG. 3R, with the etch stop elements 314, the dielectric fillings 316, and the isolation structure 114 as a self-align etching mask, one or more etching processes are used to remove the bottom portion 137c (that functions as a dummy epitaxial structure), in accordance with some embodiments. As a result, a backside contact opening 318 that exposes the remaining epitaxial structures 138s are formed. In some embodiments, no photolithography process is used for forming the backside contact opening 318.

As shown in FIGS. 3S and 4F, backside conductive contacts 322 are formed into the backside contact opening 318, in accordance with some embodiments. In some embodiments, because the isolation structure 114 is formed to have a low height, each of the backside contact openings 318 has a low aspect ratio, which facilitates the formation of the backside conductive contacts 322 in the backside contact opening 318. For example, voids are prevented from being formed in the backside contact openings 318. The reliability and conductivity of the backside contact openings 318 are improved.

Each of the backside conductive contacts 322 is electrically connected to the respective epitaxial structure 138s. Each of the backside conductive contacts 322 may function as a portion of a power rail connected to the respective epitaxial structure 138s. In some embodiments, each of the epitaxial structures 138s is wider than each of the backside conductive contacts 322, as shown in FIGS. 3S and 4F.

In some embodiments, the etch stop element 314 extends along a first sidewall of the backside conductive contact 322 and the bottom of the metal gate stack 156A or 156B, as shown in FIG. 3S. In some embodiments, the etch stop element 314 is in direct contact with the first sidewall of the backside conductive contact 322. In some embodiments, the isolation structure 114 is in direct contact with a second sidewall of the backside conductive contact 322, as shown in FIG. 4F.

In some embodiments, before the formation of the backside conductive contact 322, metal-semiconductor compound layers 320 are formed on the bottom surface of the epitaxial structures 138s that are exposed by the backside contact openings 318. Each of the metal-semiconductor compound layers 320 may be a metal silicide material that contains titanium, nickel, cobalt, tantalum, platinum, tungsten, one or more other noble metals, one or more other refractory metals, one or more other rare earth metals, one or more other suitable materials, or a combination thereof. Alternatively, the metal-semiconductor compound layer may be a metal-containing material that contains no silicon. For example, the metal-semiconductor compound layer may contain germanium and a metal element such as titanium, nickel, cobalt, tantalum, platinum, tungsten, one or more

other noble metals, one or more other refractory metals, one or more other rare earth metals, one or more other suitable materials, or a combination thereof.

In some embodiments, before the formation of the metal-semiconductor compound layer 320, the exposed epitaxial structures 138s are modified to assist in the subsequent formation of the metal-semiconductor compound layers 320. In some embodiments, one or more ion implantation processes are used to reduce the crystallinity of the surface portion of the epitaxial structures 138s, which allows a subsequently deposited metal material to react with the modified surface portions more easily. The formation of the metal-semiconductor compound layers 320 may thus be facilitated.

In some embodiments, the implantation process is a plasma doping process. Plasma may be introduced into the backside contact openings 318 to modify the exposed surface portions of the epitaxial structures 138s. In some embodiments, reaction gas used in the implantation process includes silicon-containing gas, germanium-containing gas, argon-containing gas, helium-containing gas, one or more other suitable gases, or a combination thereof.

In some embodiments, a metal-containing material is applied (or deposited) on the epitaxial structures 138s while the epitaxial structures 138s are heated, in accordance with some embodiments. In some embodiments, the metal-containing material is applied (or deposited) using a CVD process, an ALD process, a PVD process, one or more other applicable processes, or a combination thereof. Because the metal-containing material is applied during the heating of the epitaxial structure 138s, the thermal energy may help to initiate chemical reaction between the surface portions of the epitaxial structure 138s and the metal-containing material. As a result, the surface portions of the epitaxial structure 138s react with the metal-containing material, and they are transformed into the metal-semiconductor compound layer 320.

As mentioned above, the metal-containing material is applied (or deposited) on the epitaxial structures 138s while the epitaxial structures 138s are heated. In some embodiments, the epitaxial structure 138s is heated to a temperature that is in a range from about 390 degrees C. to about 440 degrees C. In some embodiments, before the metal-containing material is applied (or deposited) on the epitaxial structures 138s, the epitaxial structures 138s are heated to be at a raised temperature. Afterwards, the epitaxial structures 138s are kept at the raised temperature while the metal-containing material is applied (or deposited). The raised temperature may be in a range from about 390 degrees C. to about 440 degrees C.

Many variations and/or modifications can be made to embodiments of the disclosure. In some other embodiments, the metal-containing material is applied (or deposited) on the epitaxial structure 138s without heating the epitaxial structure 138s at the same time. A subsequent thermal operation is then used to form the metal-semiconductor compound layer 320.

In some embodiments, the metal-containing material is also applied (or deposited) on sidewalls of the backside contact opening 318 and the top surface of the dielectric fillings 316, so as to form a metal layer. Afterwards, an etching process may be used to remove the metal layer that does not react with the epitaxial structures 138s.

However, embodiments of the disclosure are not limited thereto. Many variations and/or modifications can be made

to embodiments of the disclosure. In some other embodiments, the metal-semiconductor compound layer **320** is not formed.

After the formation of the metal-semiconductor compound layer **320**, a conductive material is deposited to overfill the backside contact opening **318**, in accordance with some embodiments. The conductive material may be made of or include ruthenium, cobalt, tungsten, titanium, molybdenum, tantalum, nickel, copper, titanium nitride, tantalum nitride, one or more other suitable materials, or a combination thereof. The conductive material may be deposited using an ALD process, a CVD process, a PVD process, an electroplating process, an electroless plating process, one or more other applicable processes, or a combination thereof.

Afterwards, a planarization process is used to remove the conductive material outside of the backside contact opening **318**, in accordance with some embodiments. As a result, the remaining portions of the conductive material in the backside contact openings **318** form the backside conductive contacts **322**. The planarization process mentioned above may include a CMP process, a grinding process, an etching process, a dry polishing process, one or more other applicable processes, or a combination thereof.

As mentioned above, in some embodiments, the height ratio (H_2/H_1) of the height H_2 of the dielectric fin **204** to the height H_1 of the isolation structure **114** is in a range from about 0.75 to about 1.8. In some cases, if the height ratio (H_2/H_1) is smaller than about 0.75, the height H_1 of the isolation structure **114** may be too high, which leads to a high aspect ratio of the opening **312** or the backside contact opening **318**. The formation and quality of the dielectric filling **316** and the backside conductive contact **322** might be negatively affected. In some cases, if the height ratio (H_2/H_1) is greater than about 1.8, the height H_1 of the isolation structure **114** may be too small. The isolation structure **114** might not be able to prevent current leakage between elements such as between nearby epitaxial structures.

As shown in FIG. 3T, an interconnection structure **324** is formed over the dielectric fillings **316** and the backside conductive contact **322**, in accordance with some embodiments. Similar to the interconnection structure **160**, the interconnection structure **324** includes multiple dielectric layers and multiple conductive features including conductive lines and conductive vias. One of the conductive lines may be in electrically connected to the backside conductive contact **322**, and they may together function as a power rail.

Afterwards, the device structure is flipped upside down, and the carrier substrate **170** and the adhesion layer **162** are removed. Afterwards, an interconnection structure **326** is formed over the interconnection structure **160**, as shown in FIG. 3T in accordance with some embodiments.

In some embodiments, there are four channel structures (such as the semiconductor nanostructures **104a'-104d'**) formed. However, embodiments of the disclosure are not limited thereto. Many variations and/or modifications can be made to embodiments of the disclosure. In some embodiments, the total number of semiconductor nanostructures is greater than four. In some other embodiments, the total number of semiconductor nanostructures is smaller than four. The total number of semiconductor nanostructures (or channel structures) of each semiconductor device structure may be fine-tuned to meet requirements. For example, the total number of semiconductor nanostructures may be 3 to 8. The semiconductor nanostructures may have many appli-

cable profiles. The semiconductor nanostructures may include nanosheets, nanowires, or other suitable nanostructures.

Embodiments of the disclosure form a semiconductor device structure with a backside conductive contact and a fin structure. A stack of channel structures of the fin structures are wrapped in a metal gate stack. The isolation structure (such as a STI structure) is formed to have a low height, which enables a subsequently formed backside contact opening to have a low aspect ratio to facilitate the formation of the backside conductive contact. The reliability and quality of the semiconductor device structure are greatly improved.

In accordance with some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes a stack of channel structures and includes a first epitaxial structure and a second epitaxial structure adjacent to opposite sides of the channel structures. The semiconductor device structure also includes a gate stack wrapped around each of the channel structures and a backside conductive contact connected to the second epitaxial structure. The second epitaxial structure is between a top of the backside conductive contact and a top of the gate stack. The semiconductor device structure further includes a dielectric fin stacked over an isolation structure. The dielectric fin is adjacent to the second epitaxial structure, and the isolation structure is adjacent to the backside conductive contact. The isolation structure has a first height, the dielectric fin has a second height, and the second height is greater than the first height.

In accordance with some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes a stack of channel structures and includes a first epitaxial structure and a second epitaxial structure sandwiching the channel structures. The semiconductor device structure also includes a gate stack wrapped around the channel structures and a backside conductive contact connected to the second epitaxial structure. The backside conductive contact is below a bottom surface of the second epitaxial structure. The semiconductor device structure further includes a dielectric fin stacked over an isolation structure. The dielectric fin is adjacent to the second epitaxial structure, and the isolation structure is adjacent to the backside conductive contact. The isolation structure has a first height, the dielectric fin has a second height, and a ratio of the second height to the first height is in a range from about 0.75 to about 1.8.

In accordance with some embodiments, method for forming a semiconductor device structure is provided. The method includes forming a fin structure over a semiconductor substrate, and the fin structure has a plurality of sacrificial layers and a plurality of semiconductor layers laid out alternately. The method also includes surrounding a lower portion of the fin structure with an isolation structure. A protruding portion of the fin structure above a top of the isolation structure has a first height that is greater than a second height of the isolation structure. The method further includes forming a first epitaxial structure and a second epitaxial structure sandwiching the fin structure. In addition, the method includes removing the sacrificial layers to release a plurality of semiconductor nanostructures made of remaining portions of the semiconductor layers. The method includes forming a gate stack to wrap around the semiconductor nanostructures. The method also includes forming a backside conductive contact connected to the second epitaxial structure. The backside conductive contact is adjacent to the isolation structure.

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The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device structure, comprising:
 - a stack of channel structures;
 - a first epitaxial structure and a second epitaxial structure adjacent to opposite sides of the channel structures;
 - a gate stack wrapped around the channel structures;
 - a backside conductive contact connected to the second epitaxial structure, wherein the second epitaxial structure is between a top of the backside conductive contact and a top of the gate stack; and
 - an etch stop layer extending along a sidewall of the backside conductive contact and a bottom of the gate stack.
2. The semiconductor device structure as claimed in claim 1, further comprising a metal-semiconductor compound layer between the backside conductive contact and the second epitaxial structure, wherein the metal-semiconductor compound layer is in direct contact with the etch stop layer.
3. The semiconductor device structure as claimed in claim 1, wherein the etch stop layer contains nitrogen.
4. The semiconductor device structure as claimed in claim 1, wherein the second epitaxial structure is wider than the backside conductive contact.
5. The semiconductor device structure as claimed in claim 1, further comprising:
 - a dielectric filling laterally surrounding the backside conductive contact, wherein the etch stop layer is between the backside conductive contact and the dielectric filling.
6. The semiconductor device structure as claimed in claim 5, further comprising:
 - an isolation structure laterally surrounding the dielectric filling and the backside conductive contact.
7. The semiconductor device structure as claimed in claim 6, wherein the backside conductive contact is in direct contact with the isolation structure.
8. The semiconductor device structure as claimed in claim 6, wherein the dielectric filling is separated from the isolation structure by the etch stop layer.
9. The semiconductor device structure as claimed in claim 6, wherein the dielectric filling extends past interfaces between the first epitaxial structure and the channel structures.
10. The semiconductor device structure as claimed in claim 6, wherein the backside conductive contact is in direct contact with the isolation structure and the etch stop layer.
11. A semiconductor device structure, comprising:
 - a stack of channel structures;

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- a first epitaxial structure and a second epitaxial structure sandwiching the channel structures;
 - a gate stack wrapped around the channel structures;
 - a backside conductive contact connected to the second epitaxial structure, wherein the backside conductive contact is below a bottom surface of the second epitaxial structure;
 - a dielectric filling laterally surrounding the backside conductive contact; and
 - an isolation structure laterally surrounding the dielectric filling and the backside conductive contact.
12. The semiconductor device structure as claimed in claim 11, wherein the backside conductive contact is in direct contact with the isolation structure.
 13. The semiconductor device structure as claimed in claim 11, further comprising:
 - a nitride layer between the dielectric filling and the backside conductive contact.
 14. The semiconductor device structure as claimed in claim 13, wherein the backside conductive contact is in direct contact with the isolation structure and the nitride layer.
 15. The semiconductor device structure as claimed in claim 11, wherein the second epitaxial structure extends past opposite edges of the backside conductive contact.
 16. A semiconductor device structure, comprising:
 - a stack of channel structures;
 - a first epitaxial structure and a second epitaxial structure sandwiching the channel structures;
 - a gate stack wrapped around the channel structures;
 - a backside conductive contact connected to the second epitaxial structure, wherein the backside conductive contact is below a bottom surface of the second epitaxial structure, and the second epitaxial structure extends past opposite edges of the backside conductive contact; and
 - a dielectric filling laterally surrounding the backside conductive contact, wherein the first epitaxial structure extends past opposite edges of the dielectric filling.
 17. The semiconductor device structure as claimed in claim 16, further comprising:
 - an isolation structure laterally surrounding the backside conductive contact and the dielectric filling.
 18. The semiconductor device structure as claimed in claim 17, wherein the backside conductive contact is in direct contact with the isolation structure.
 19. The semiconductor device structure as claimed in claim 17, further comprising:
 - a first dielectric fin and a second dielectric fin stacked over the isolation structure, wherein the second epitaxial structure is between the first dielectric fin and the second dielectric fin.
 20. The semiconductor device structure as claimed in claim 19, wherein the first dielectric fin is taller than the backside conductive contact.

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